

# Digital Psychiatrist

AI Mental Health Assistant



Smart  
Conversation



Patient  
Support



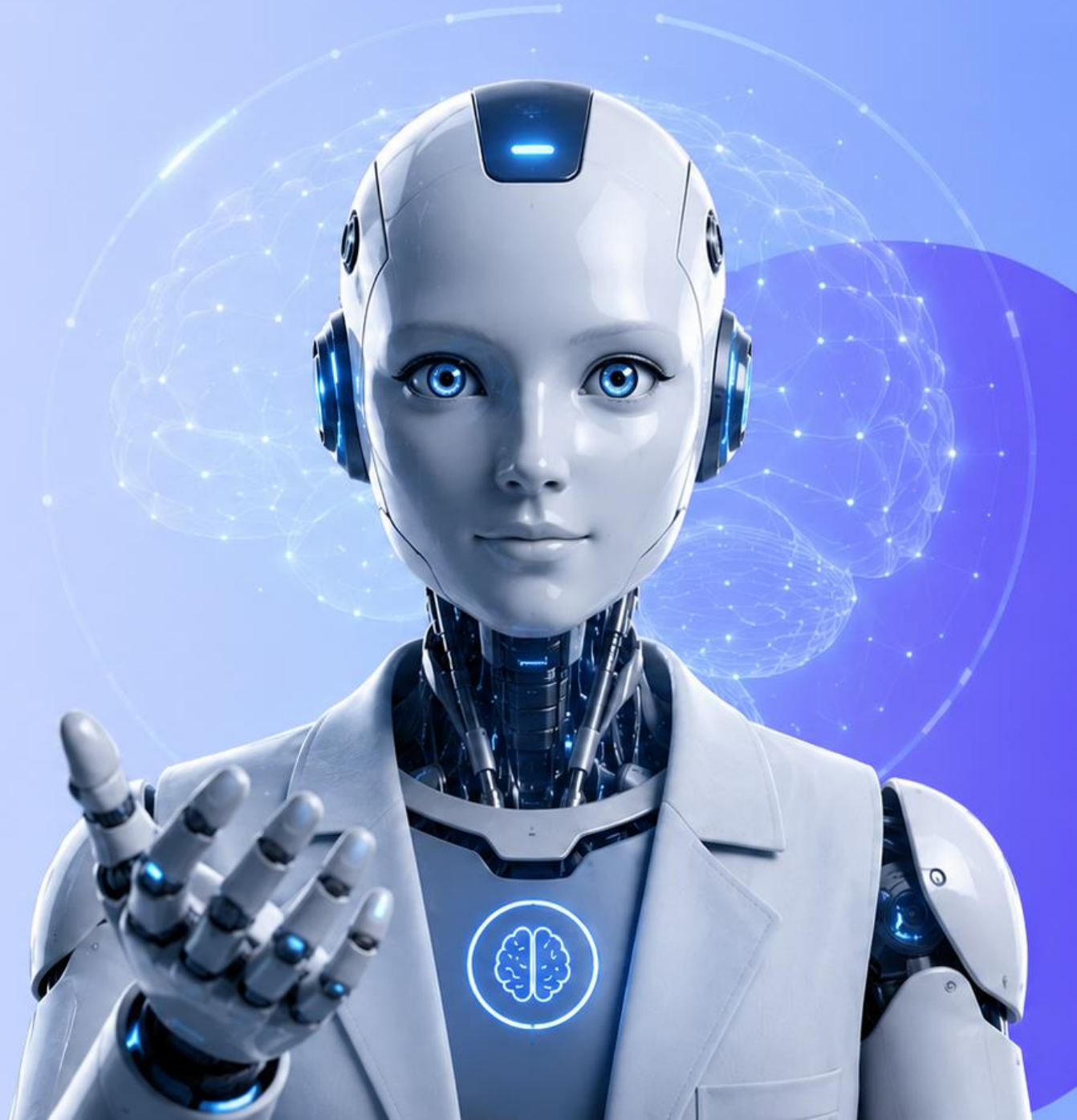
Mental Health  
Analysis



Psychiatric  
Evaluation

## Supervisors

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Eng. Khaled Mohamed



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# Introduction

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Smart solution for better mental health

- ❖ **Mental health impacts every aspect of daily life.**  
It affects how people think, feel, and interact with others.
- ❖ **The demand for mental health support continues to grow**  
More people are seeking help than ever before.
- ❖ **Digital solutions can make support more accessible**  
Providing guidance anytime, anywhere, for those who need it.



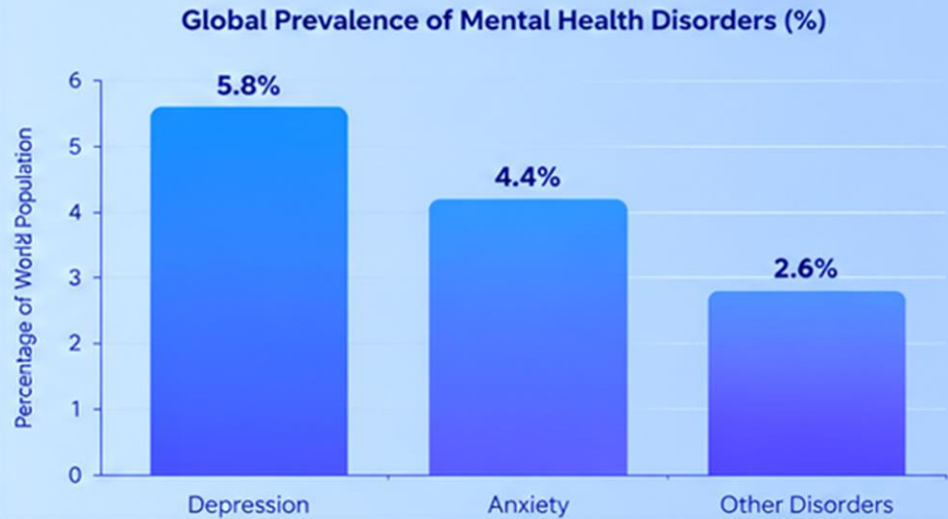
people worldwide live with a mental health condition.



Technology can provide easier access to support.



The need for mental health support is growing rapidly.







# Digital Mental Health Market



## Global Overview

### Growing Mental Health Awareness

More people are seeking mental health support worldwide.

### Rise of Digital Therapy

More People Use Online Therapy.

### AI-Powered Healthcare

AI makes mental health support more accessible.

### Expanding Market Demand

The digital mental health market continues to grow rapidly.



## Egypt Overview



### Rising Mental Health Awareness

Growing Mental Health Awareness.



### Strong Digital Connectivity

High internet usage supports digital healthcare solutions.



### Gap in Mental Healthcare Access

Limited Access to Mental Healthcare



### Growing Adoption of Digital Health

Online healthcare services are becoming increasingly accepted

# Key challenges



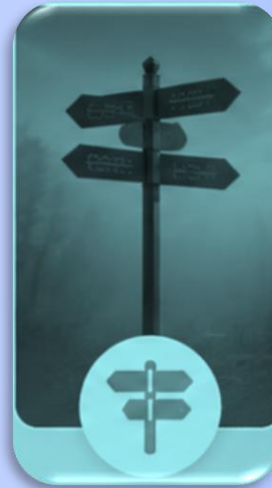
**Mental Health  
Struggles**



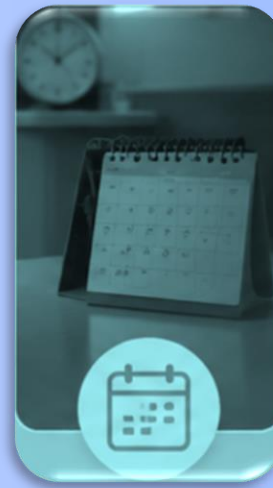
**Fear of  
Judgment**



**Fear of  
Judgment**



**Finding the  
Right Therapist**



**Complicated  
Booking**



**Long Waiting  
Times**



**Overcrowded  
Clinics**



**Mental healthcare should be  
accessible, private, and available when needed.**



# Our Solution



## Virtual Assistant

Conducts an initial psychological Assessment to understand the patient's condition



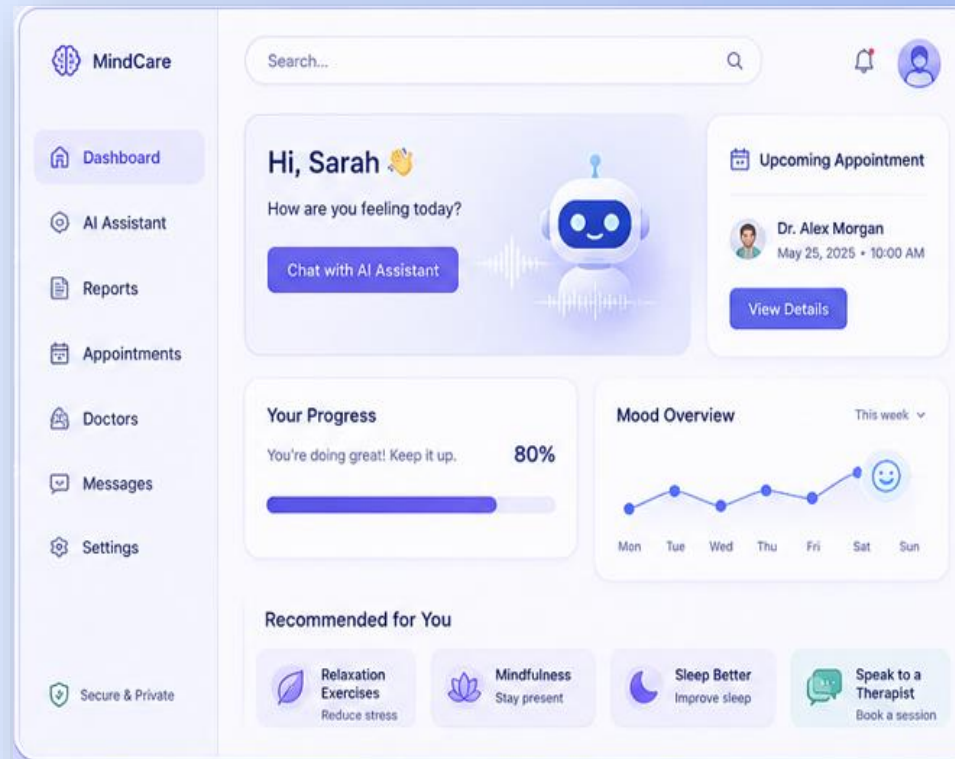
## AI Chatbot

Provides instant interaction, answers questions, and collects information about the patient's mental well-being.



## Doctor Profiles

Stores chat records and consultation reports securely for future access and follow-up.



## Easy Booking

Allows patients to schedule appointments quickly, either online or at the clinic.



## Secure History

Stores chat records and consultation reports securely for future access and follow-up.



## Secure Access

Provides role-based permissions to ensure secure access for patients, doctors, and administrators.





# Literature review



| Feature                 | BetterHelp | TalkSpace | Youper | Wysa | Woebot | MindDoc | Our Idea |
|-------------------------|------------|-----------|--------|------|--------|---------|----------|
| AI Robot                | ✗          | ✗         | ✗      | ✗    | ✗      | ✗       | ✓        |
| AI Chatbot              | ✗          | ✗         | ✓      | ✓    | ✓      | ✓       | ✓        |
| Appointment Booking     | ✓          | ✓         | ✗      | ✗    | ✗      | ✗       | ✓        |
| Doctor Recommendation   | ✗          | ✗         | ✗      | ✗    | ✗      | ✗       | ✓        |
| Digital Prescription    | ✗          | ✗         | ✗      | ✗    | ✗      | ✗       | ✓        |
| Doctor Rating           | ✗          | ✓         | ✗      | ✗    | ✗      | ✗       | ✓        |
| Doctor Profile          | ✗          | ✓         | ✗      | ✗    | ✗      | ✗       | ✓        |
| Online Therapy Sessions | ✓          | ✓         | ✗      | ✗    | ✗      | ✗       | ✓        |
| AI + Therapists         | ✗          | ✗         | ✗      | ✗    | ✗      | ✗       | ✓        |



PixVerse.ai





# Model Accuracy





01

02

03

04





## Source

Distress Analysis  
Interview Corpus  
(DAIC-WOZ)  
Clinical dataset for  
mental health research

02

03

04





# DAIC-WOZ

## Dataset

### Source

Distress Analysis  
Interview Corpus  
(DAIC-WOZ)  
Clinical dataset for  
mental health research

### Content

189 multi-modal clinical  
interviews between  
virtual agent (Ellie) and  
human participants  
Includes audio features +  
text transcripts

03

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### Clinical Labeling

Gold-standard scale for  
depression | Score  
range :0-24  
Threshold: score  $\geq 10$   
moderate-to-severe  
depression







# DAIC-WOZ

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### Reference

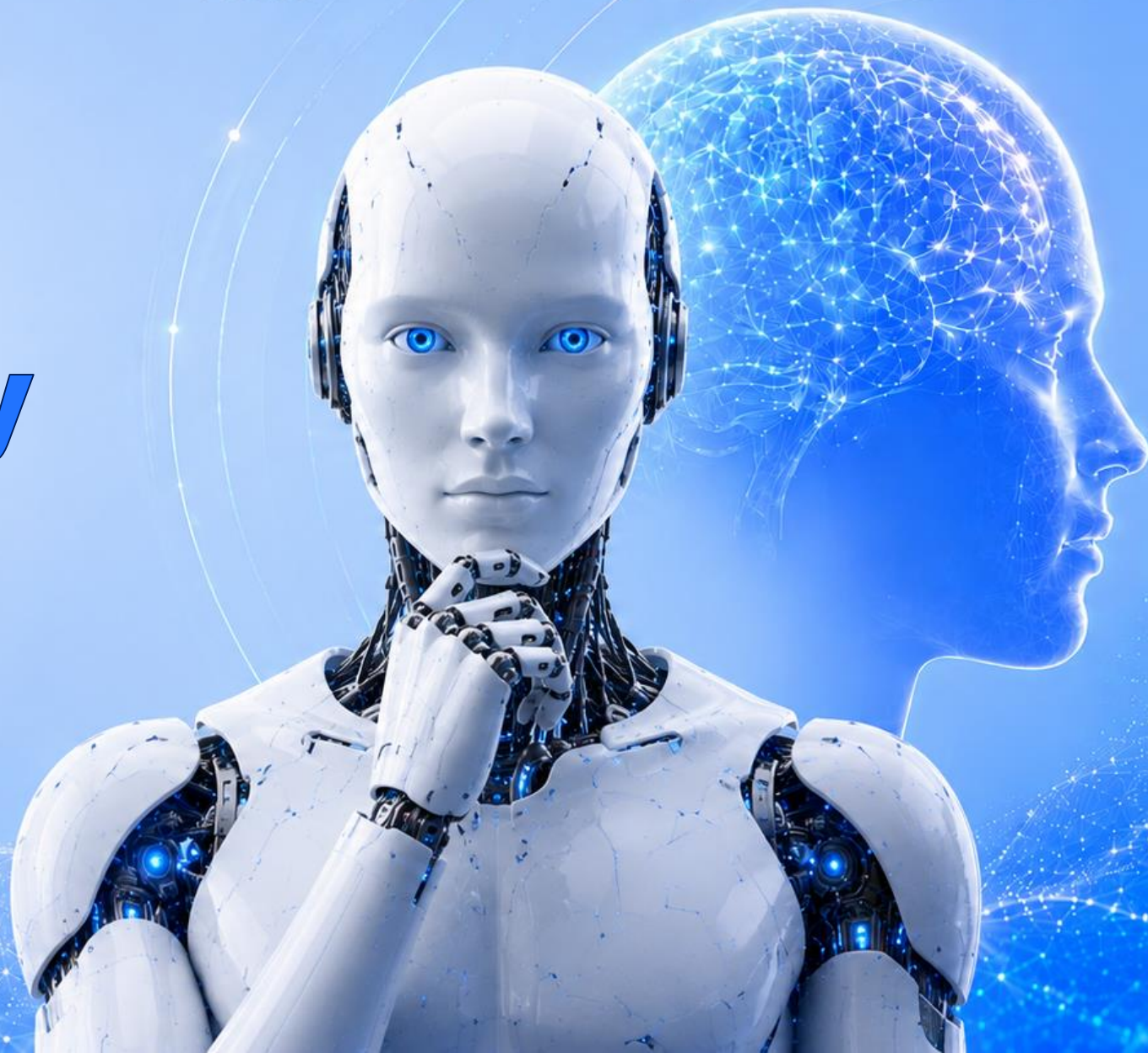
USC Institute for Creative  
Technologies

Link

<https://dcapswoz.ict.usc.edu/>



# **Text Accuracy**





# NLP Pipeline & Model Optimization



NLP PHQ-8

## Goal

Develop a hybrid NLP-based AI system to predict continuous PHQ-8 scores (0–24) and dynamically assess depression severity.



## Data Augmentation

Synthetic Extreme Cases (Healthy=0, Severe=24)

Forces clear decision boundaries and teaches the model definitive clinical keywords.



## Text Preprocessing

TF-IDF Vectorizer (max\_features=2000, 1–2 n-grams)

Extracts high-impact emotional terms while filtering conversational noise (stopwords).



## Best Model: Ridge Regression

L2 Regularization Pipeline

Prevents overfitting in high-dimensional sparse text data, ensuring stability on unseen patients.



## Core Function & Efficiency

Score  $\geq 10 \rightarrow$  Binary Classification

Predicts continuous scores for severity, then classifies (Depressed / Non-depressed).

Lightweight for real-time chatbot API integration.



## Algorithm Benchmarking

Linear · Ensemble · SVM Models

Extensive evaluation: Ridge, Lasso, Random Forest, XGBoost, and SVMs.



## Evaluation Metrics

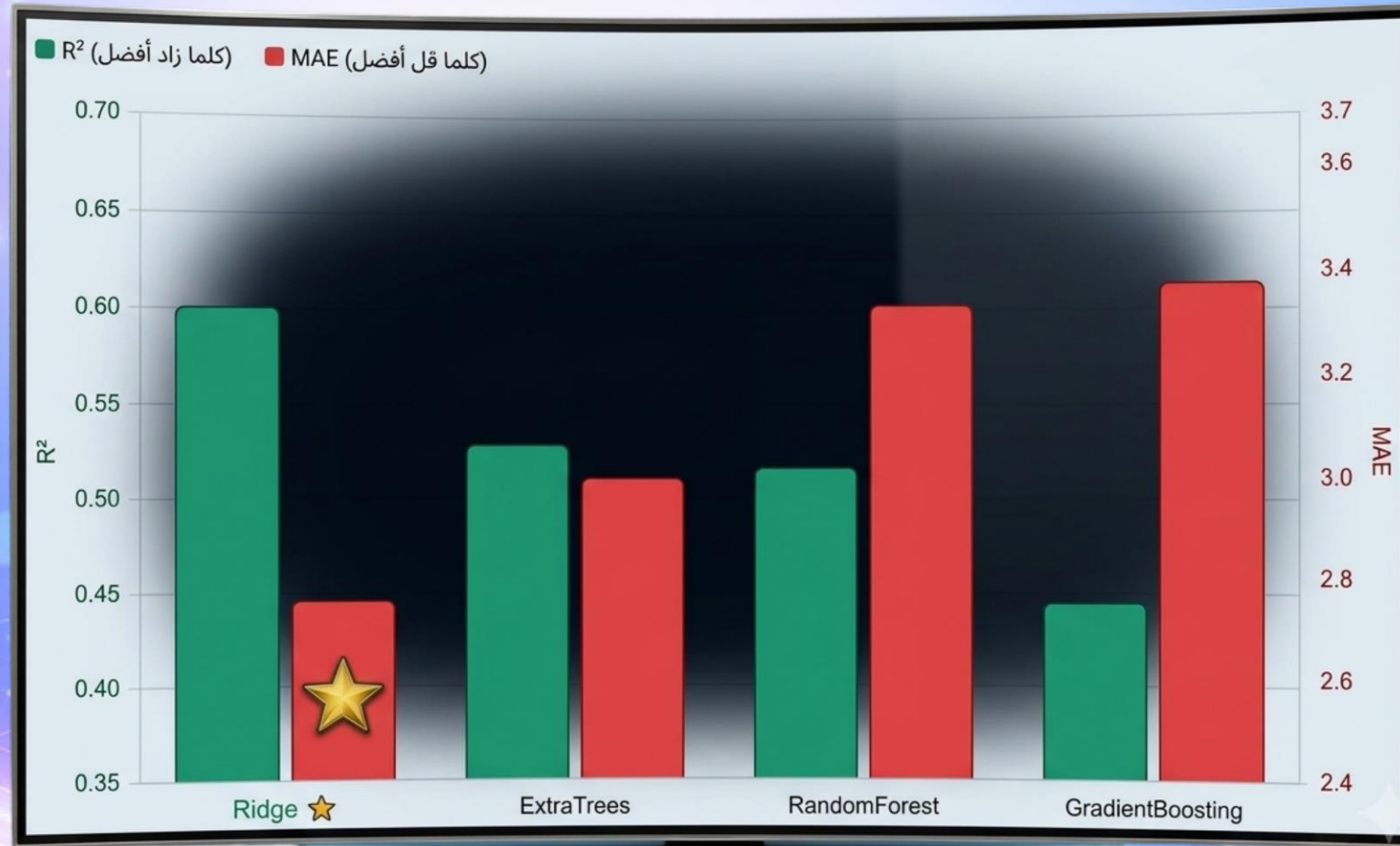
Regression + Diagnostic

$R^2$  (fit), MAE (error margin) · Accuracy, Precision, Recall, F1-score, Confusion Matrix.

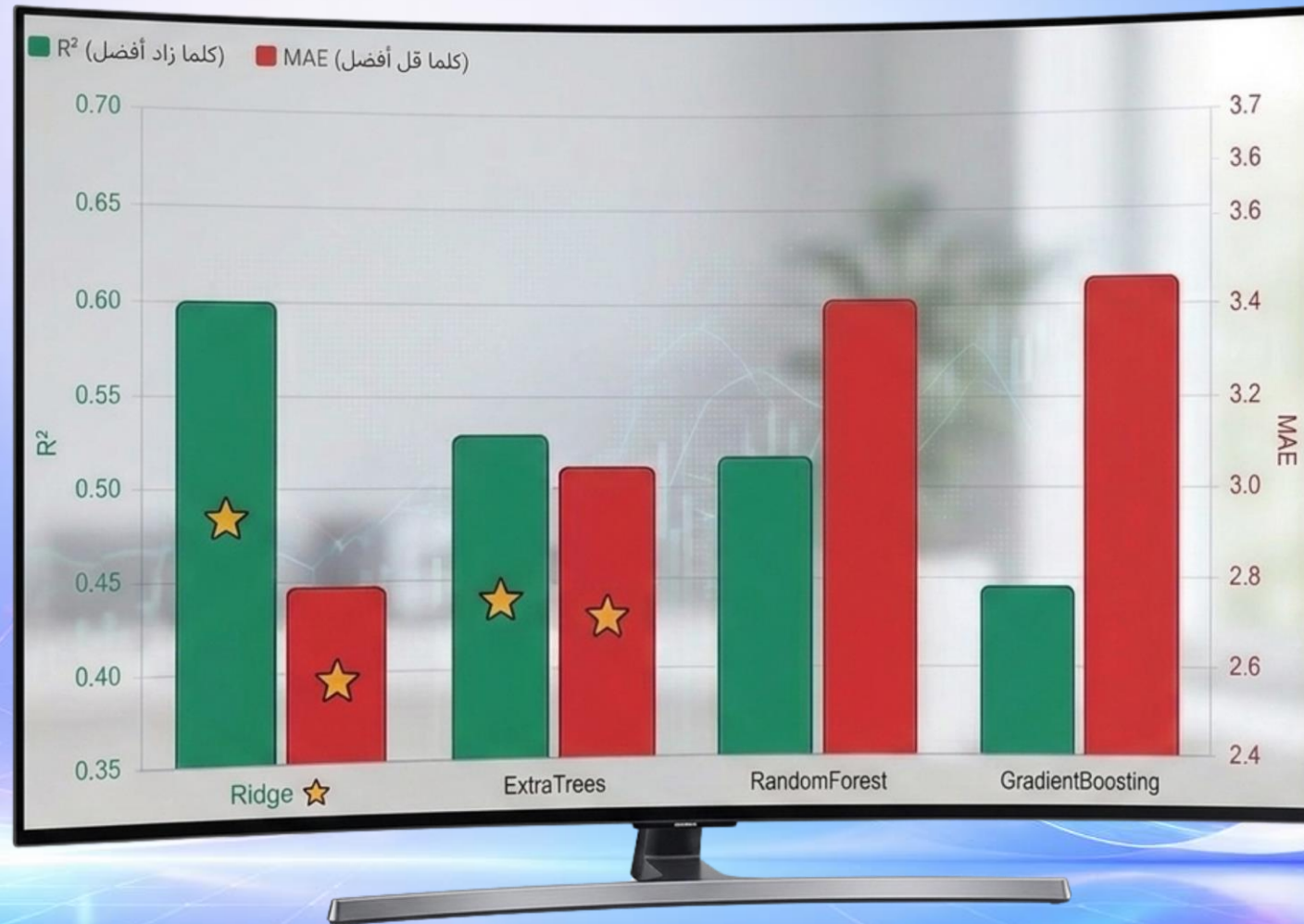




# NLP Pipeline & Model Optimization

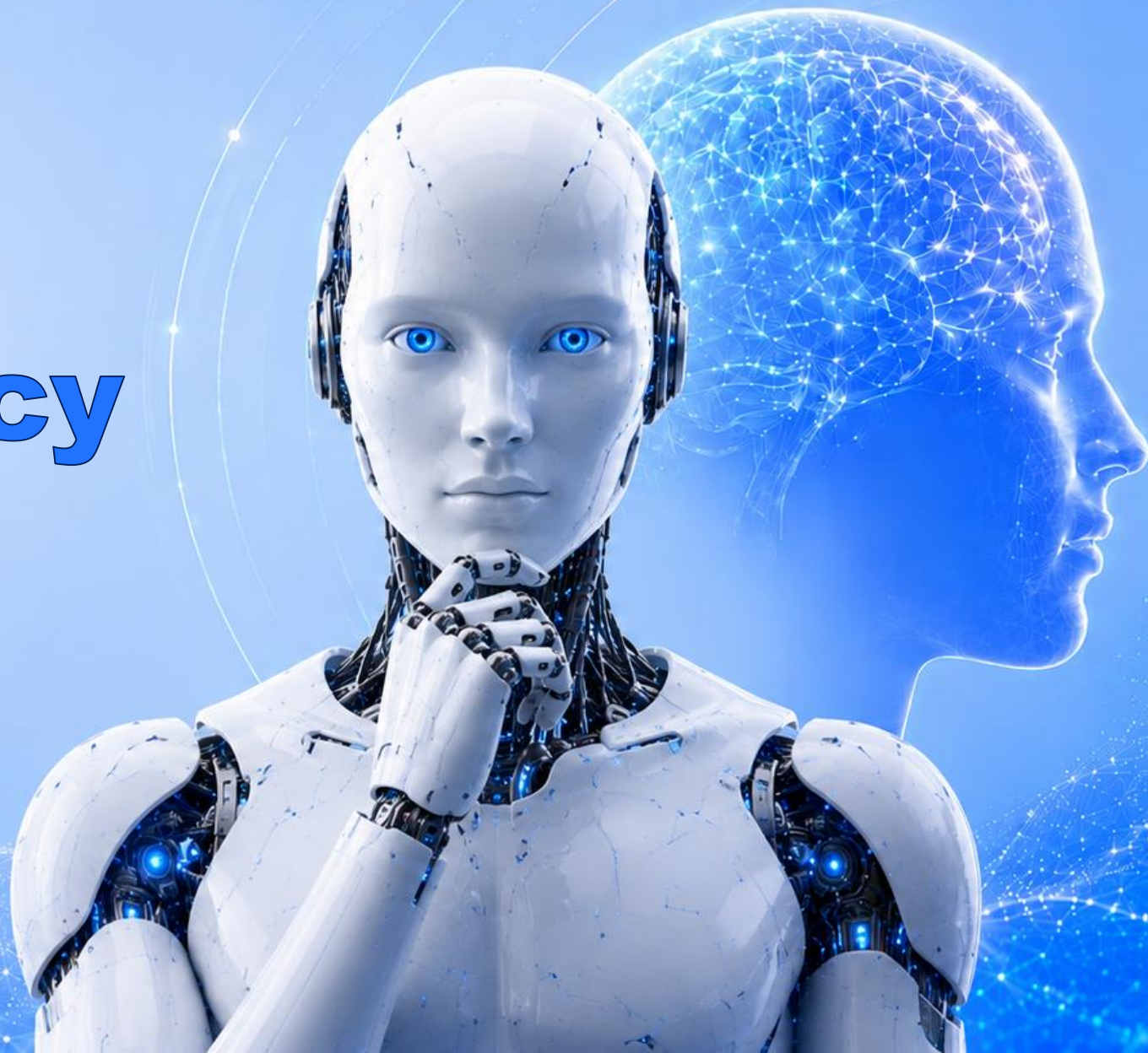


# NLP Pipeline & Model Optimization





# Audio Accuracy



**Model Objective:** Develop machine learning models to classify depression and predict PHQ-8 scores.

**Data Processing:**

- Preprocessed speech and text features .
- Applied feature normalization .
- Selected top 25 features for Classification .
- Selected top 4 features for Regression .
- Balanced the training data .
- Split the dataset into training 80 % and testing 20%

**Best Models:**

- Extra Trees Classifier
- Ridge Regression

**Algorithms Tested:**

**Classification**

- Extra Trees
- Random Forest
- SVM

**Regression**

- Ridge Regression
- Random Forest Regressor
- Gradient Boosting Regressor
- SVR

**Evaluation Metrics:**

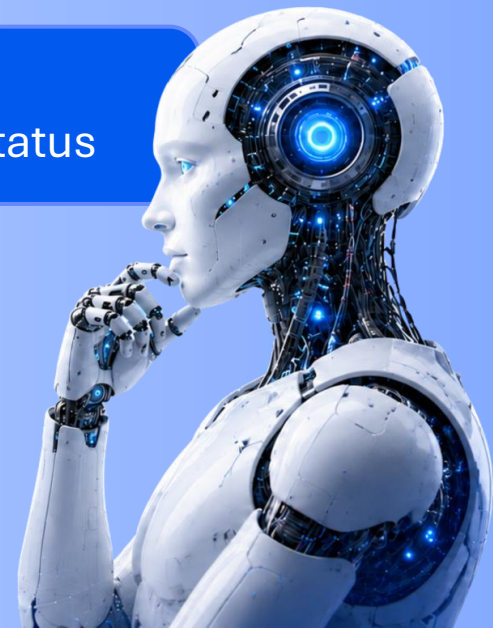
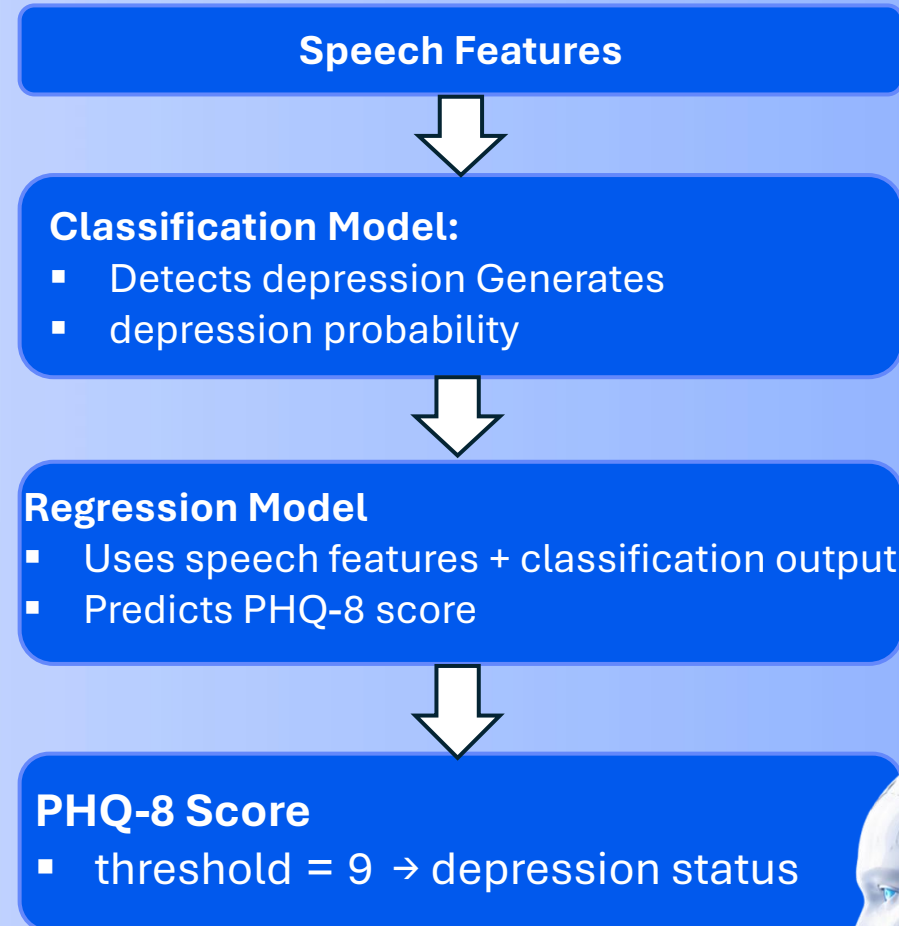
**Classification**

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

**Regression**

- $R^2$  Score
- MAE

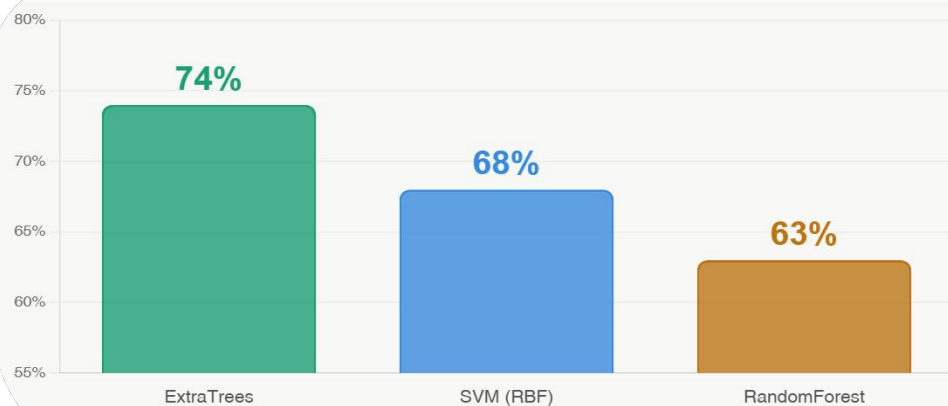
**Prediction Process**





## Direct Classification

Accuracy of detecting depression directly



## Regression Performance

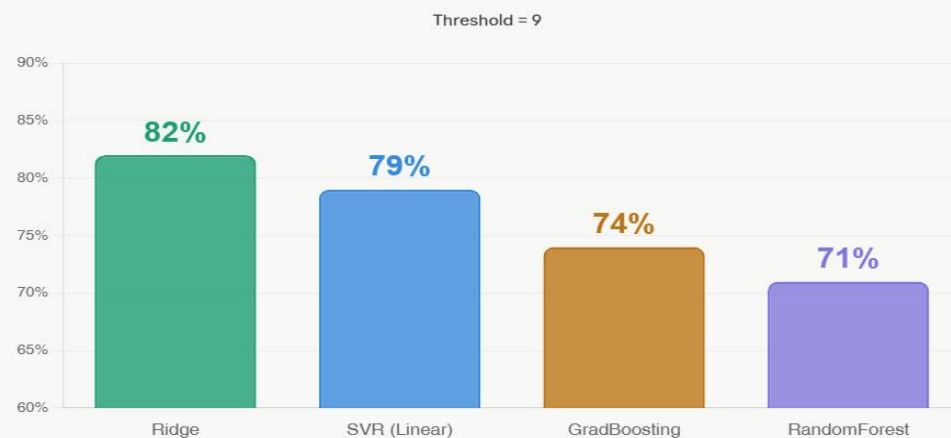
Predicting the PHQ-8 score ( $R^2$  and MAE).



Best

## Integrated Synergy

Applying Threshold = 9 to Regression outputs.



# Hybrid Text & Audio





# Multimodal Depression Detection

## Text + Audio Fusion System

### System Overview

- Combines patient text and audio recordings as inputs
- Text pipeline: TF-IDF features fed into Ridge regression
- Audio pipeline: ExtraTrees classifier + Hybrid Ridge regression
- Fusion layer:  $0.62 \times \text{text score} + 0.38 \times \text{audio score}$
- Detection threshold: PHQ  $\geq 9$  indicates depression

Text only

**83%**

R<sup>2</sup>=0.37  
MAE=3.30

Audio only

**82%**

R<sup>2</sup>=0.60  
MAE=2.65

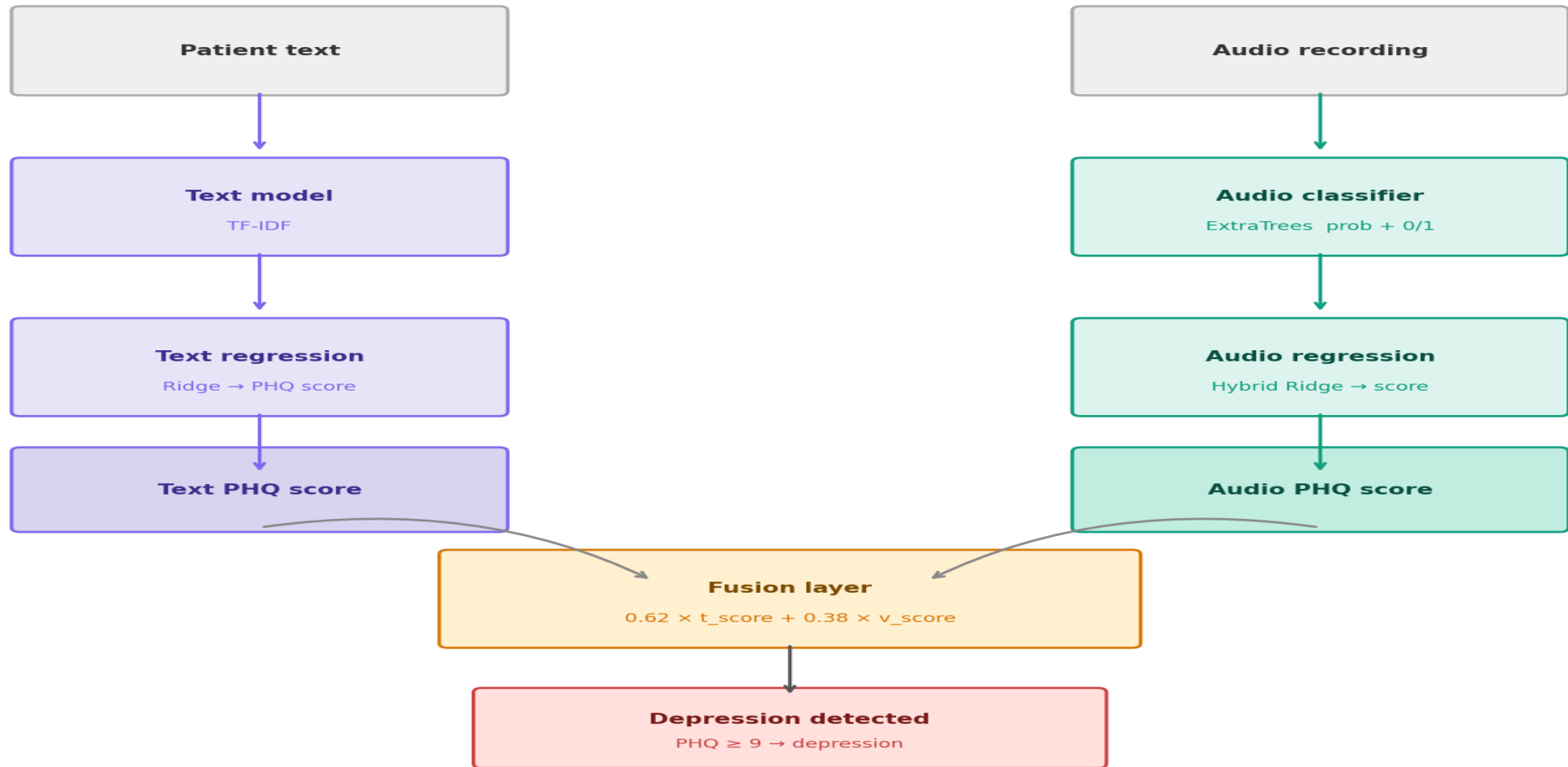
Hybrid Fusion

**91%**

R<sup>2</sup>=0.76  
MAE=2.18

# Pipeline Architecture

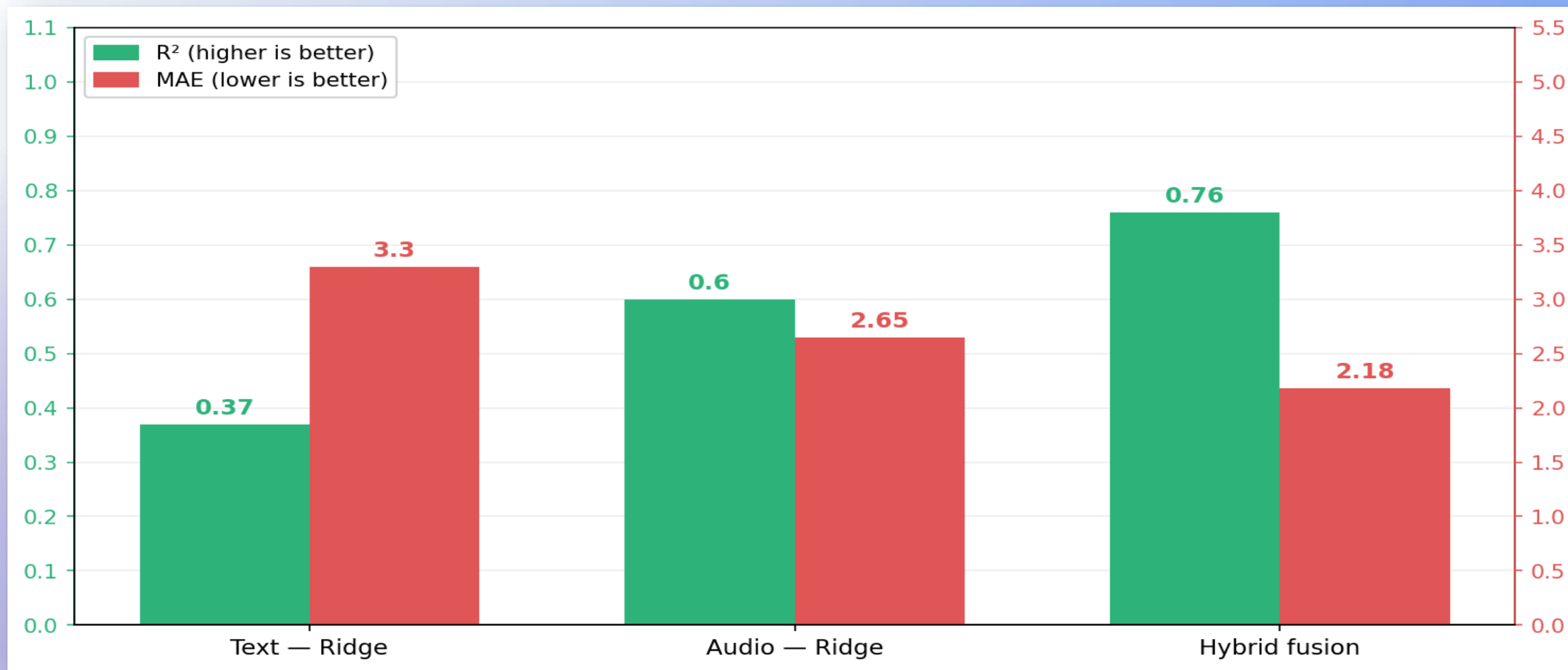
Text and audio streams processed separately, then merged via a weighted fusion layer





# Regression Performance

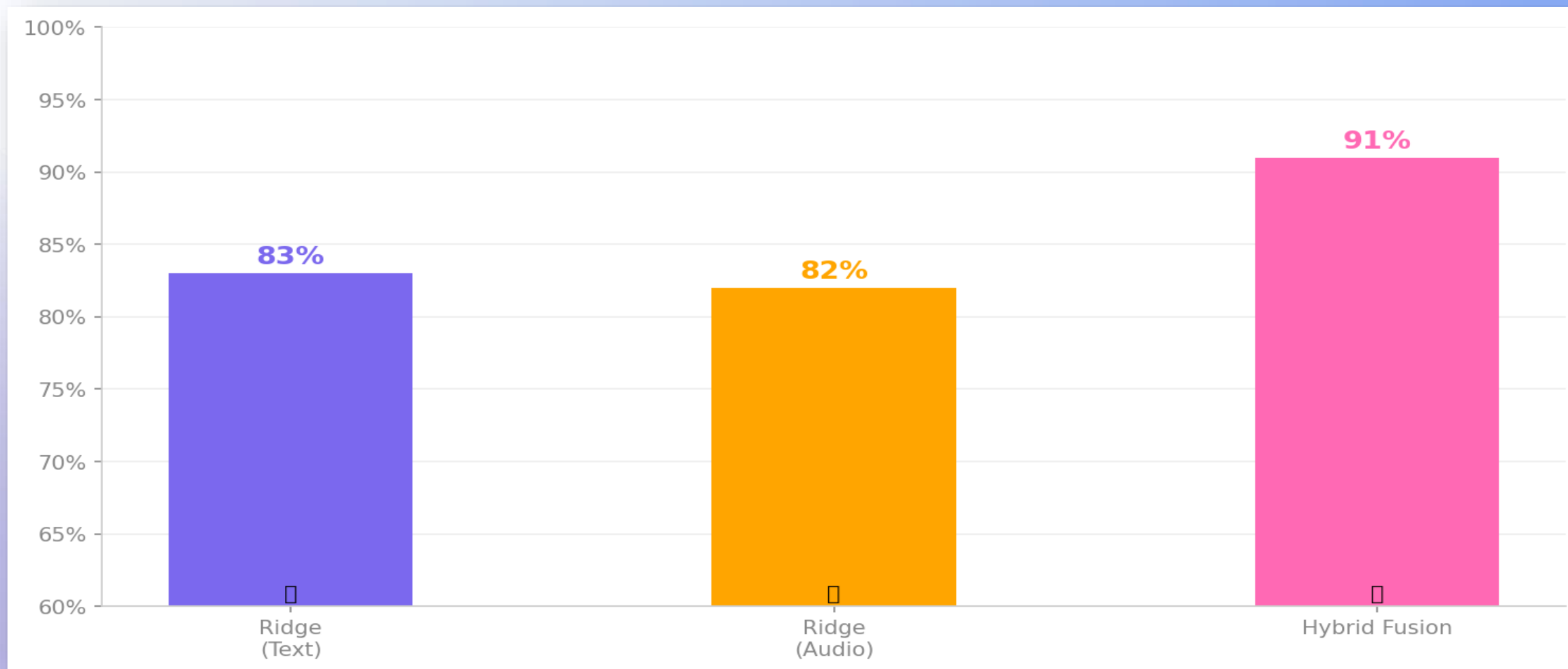
$R^2$  (higher is better) vs MAE (lower is better) — across all three model variants



Hybrid Fusion:  $R^2=0.76$  (best) | MAE=2.18 (lowest error) -> strongest overall predictor

# Classification Accuracy

Depression detection accuracy per modality — PHQ  $\geq 9$  classification threshold



Hybrid Fusion achieves 91% accuracy — +8pp over text-only, +9pp over audio-only





# System Analysis

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# Use Case Diagram

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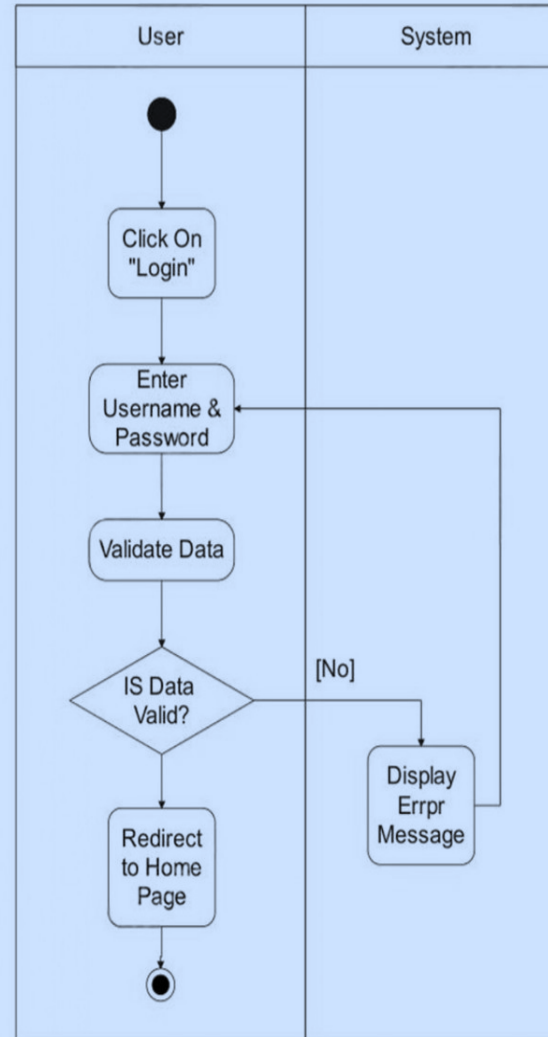
# Activity Diagram

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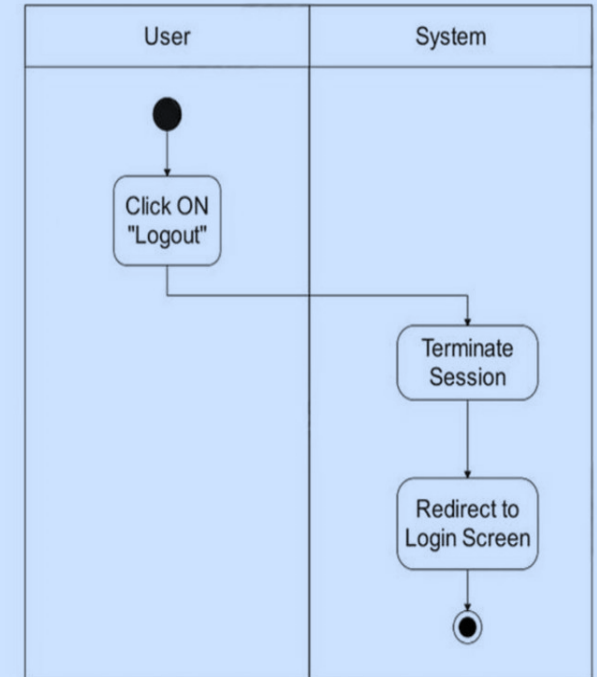
Smart solution for better mental health



Activity  
Diagram For  
Use Case :  
(Login)

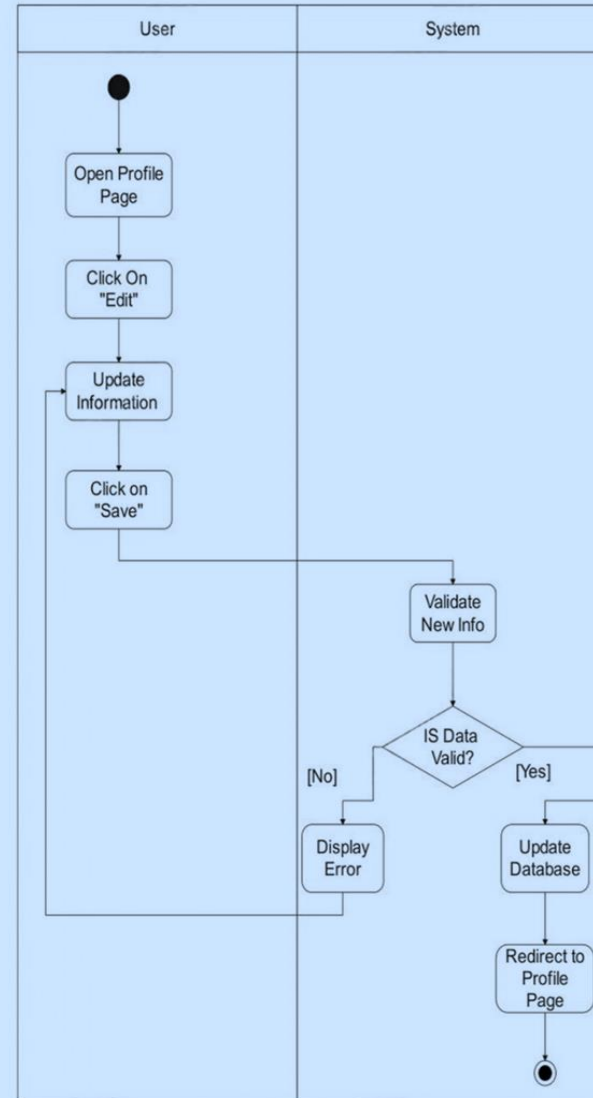


Activity Diagram  
For Use Case :  
(Logout)

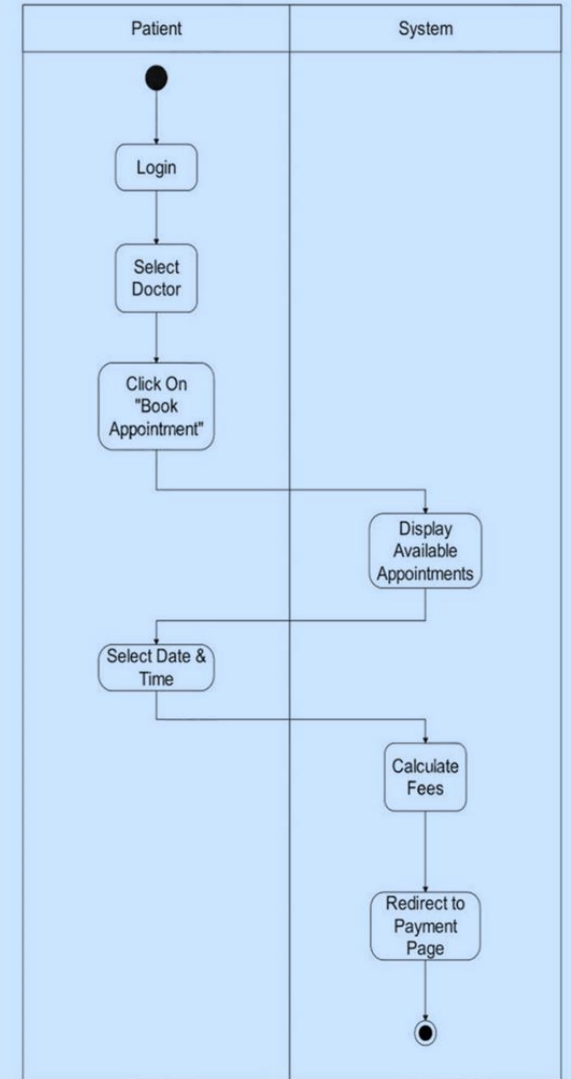




Activity  
Diagram For  
Use Case :  
(Edit Profile)



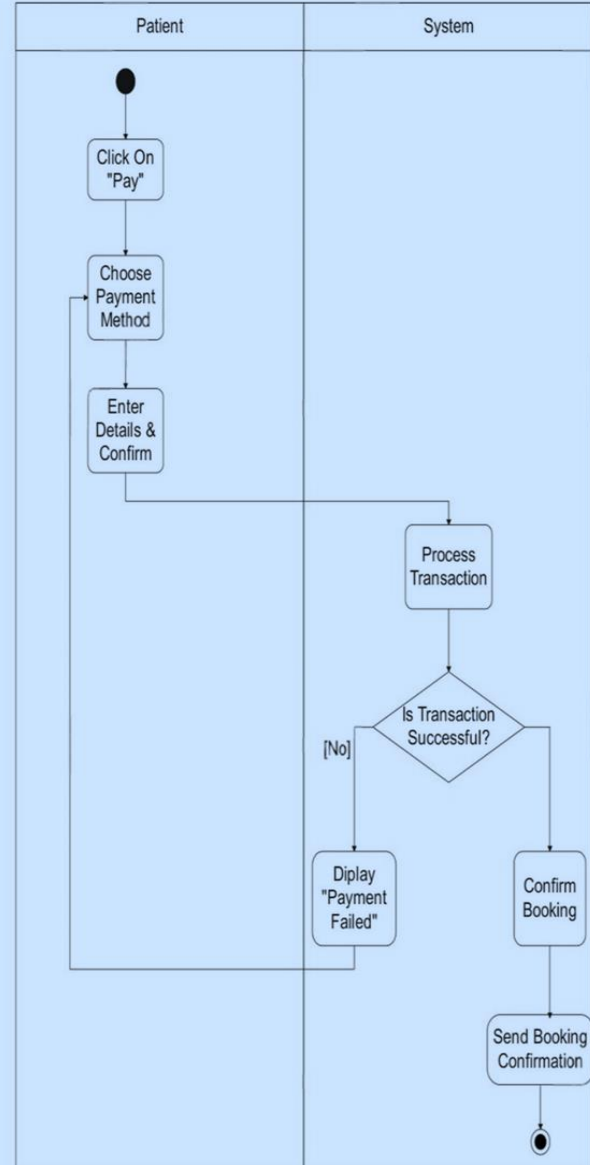
Activity Diagram  
For Use Case :  
(Book Appointment)



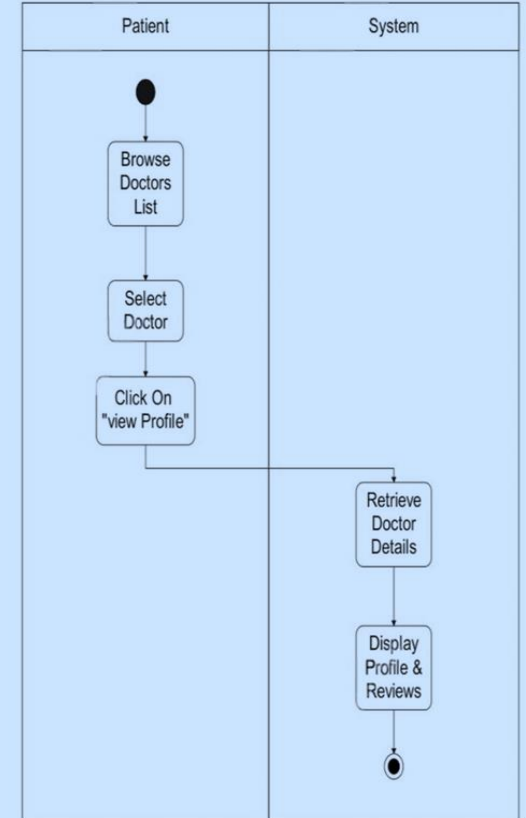




Activity Diagram For  
Use Case : (Pay Online)

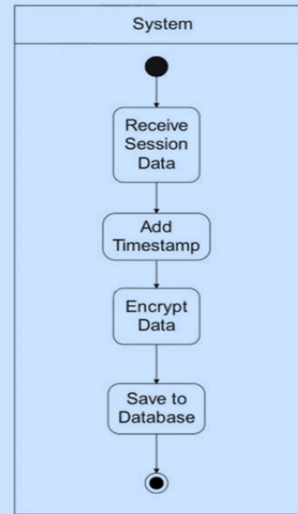


Activity Diagram For Use  
Case : (View Doctor Profile)

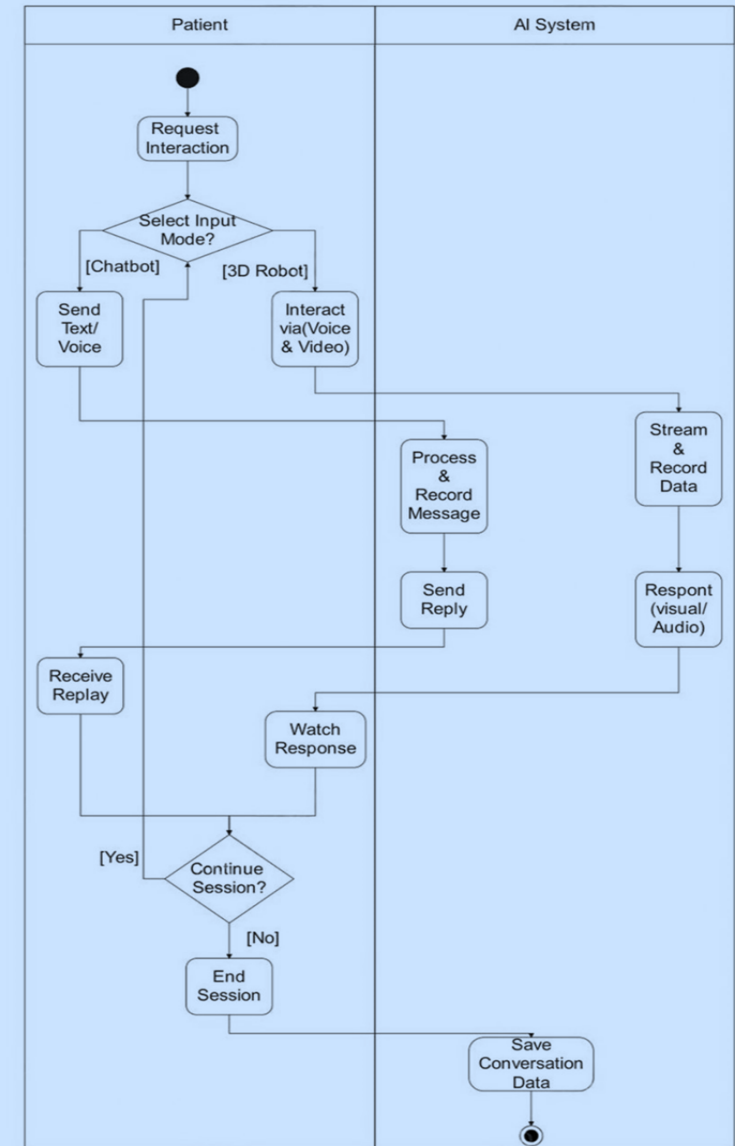




Activity Diagram For Use Case : (Log Interaction)

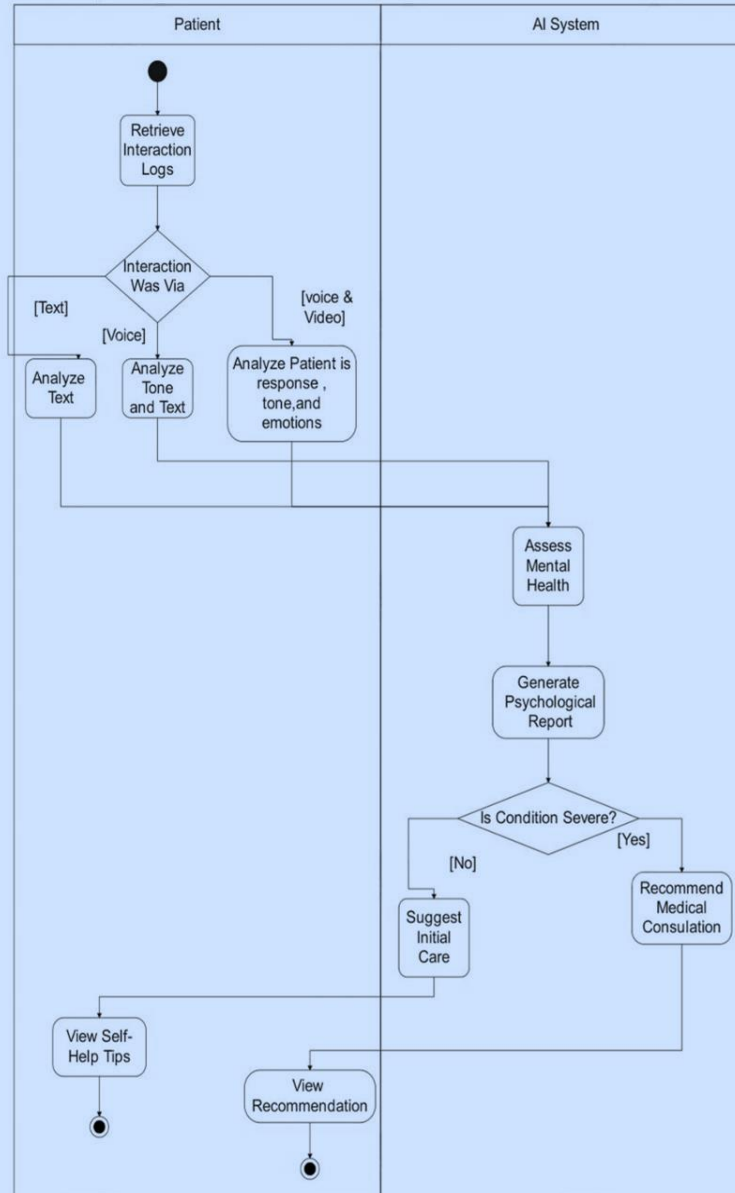


Activity Diagram For Use Case : (Interact With Chatbot)

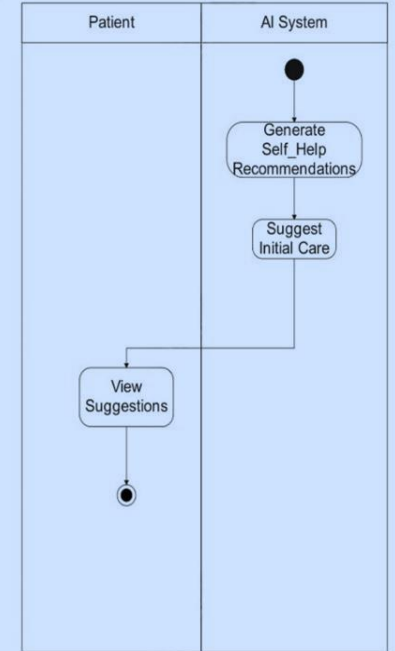




Activity Diagram For  
Use Case : (Assess  
Mental Health)



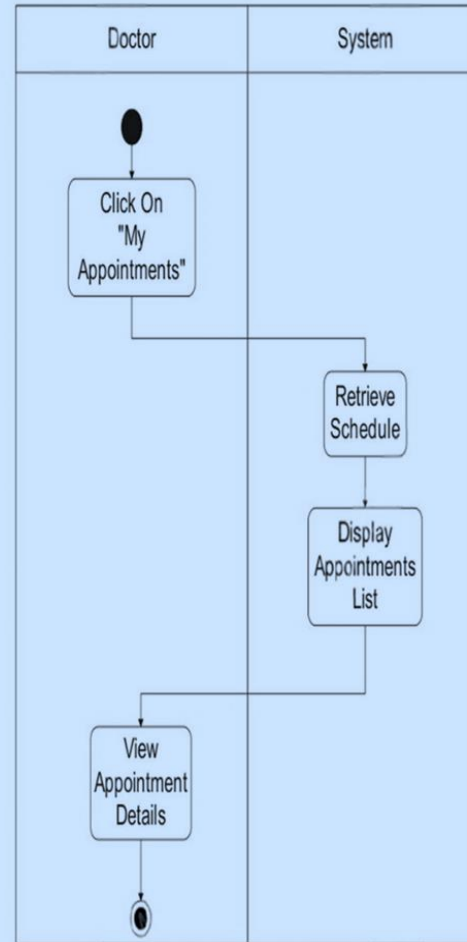
Activity Diagram For  
Use Case : (Suggest  
Initial Care)



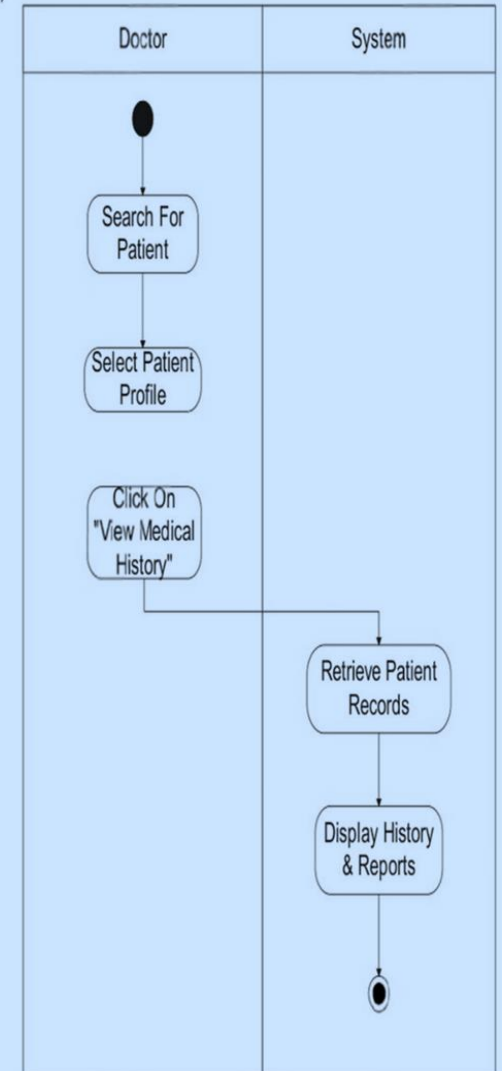




Activity Diagram  
For Use Case :  
(Review  
Appointments)

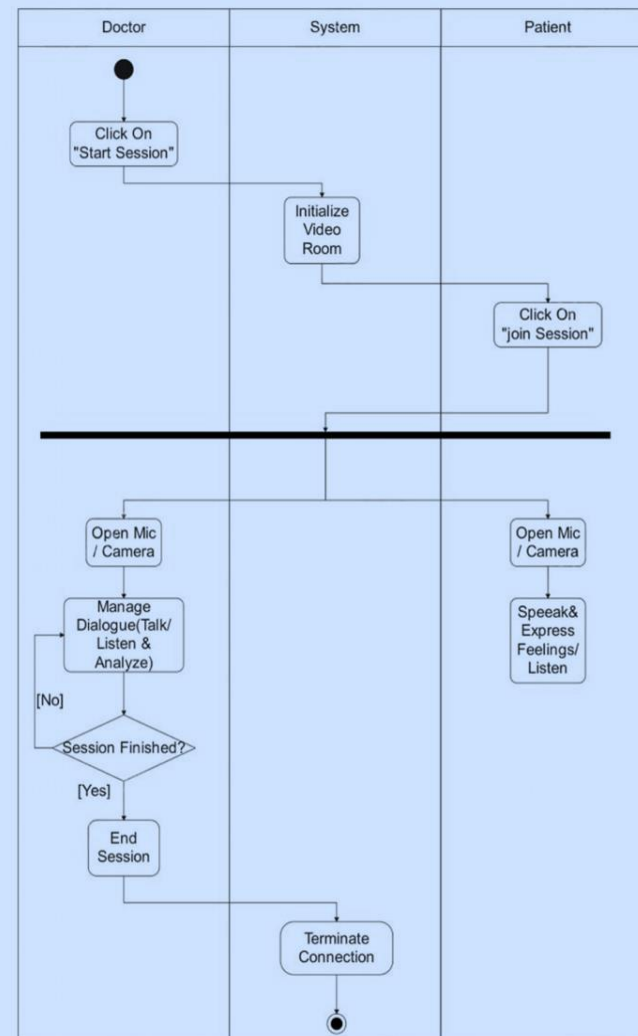


Activity Diagram For  
Use Case :  
(View Patient  
History)

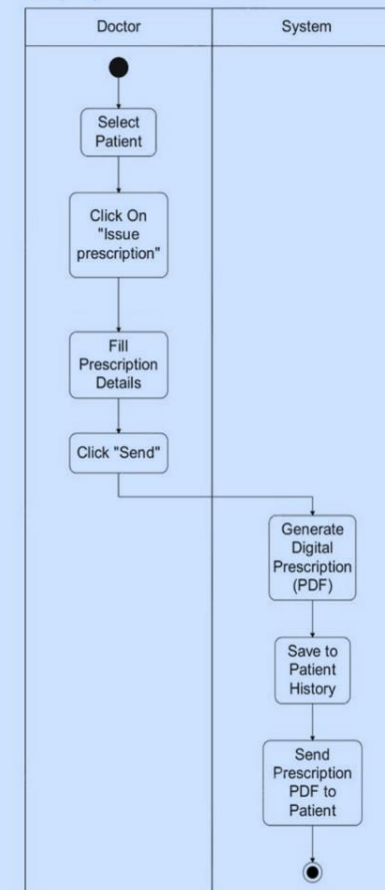




Activity  
Diagram For  
Use Case :  
(Communicate  
With Patients)

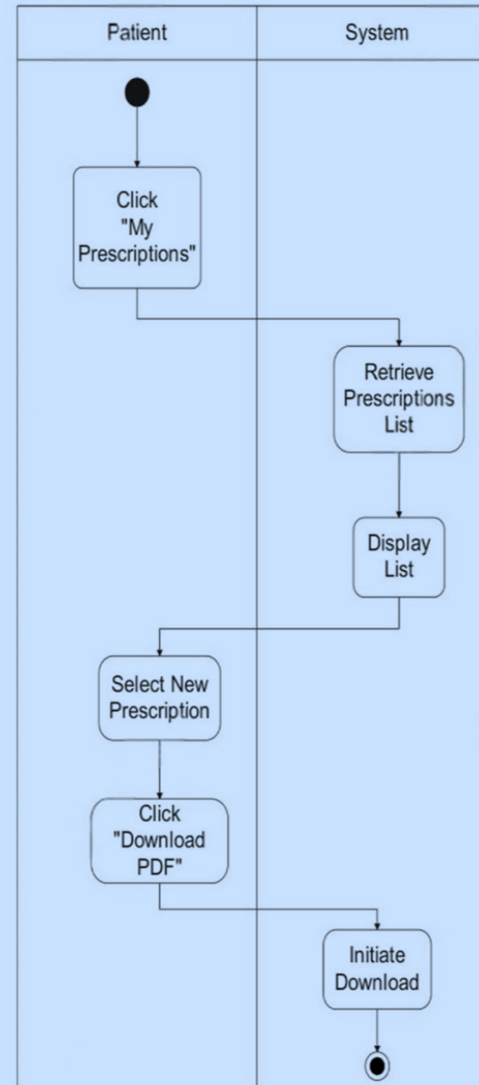


Activity Diagram  
For Use Case :  
(Send  
Prescription)

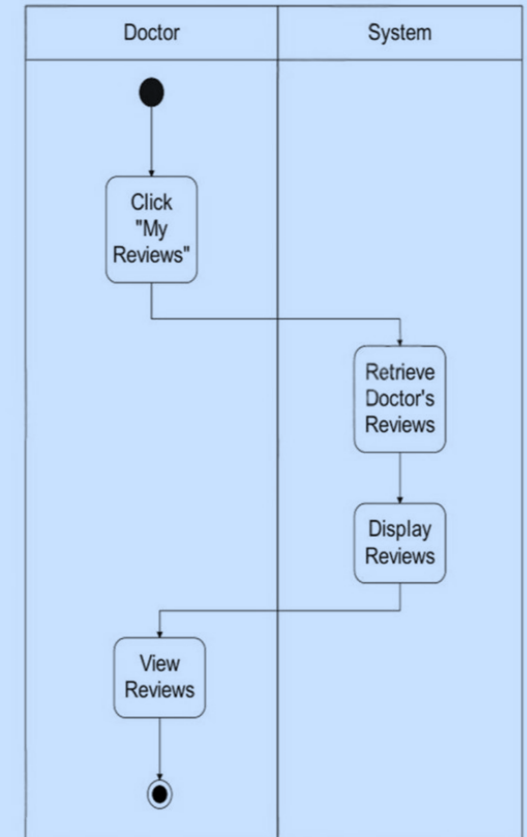




Activity Diagram For  
Use Case :  
(Receive Prescription)



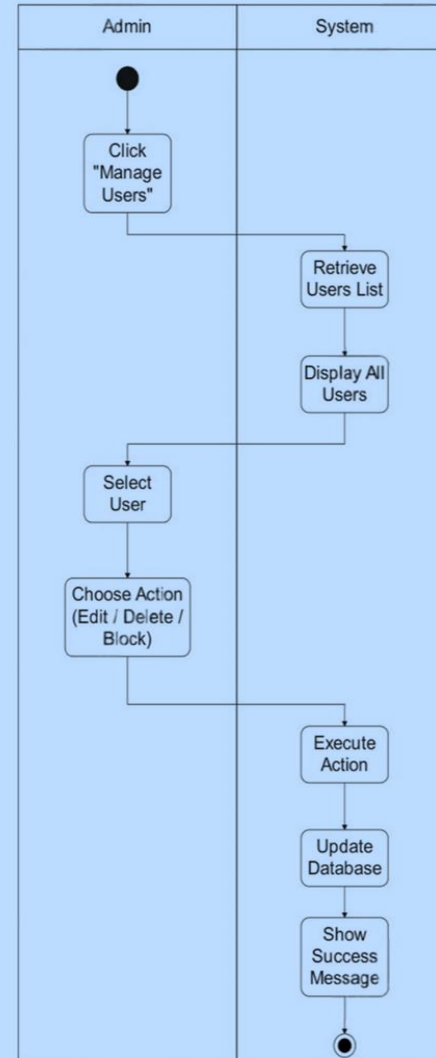
Activity Diagram For  
Use Case :  
(View Reviews)



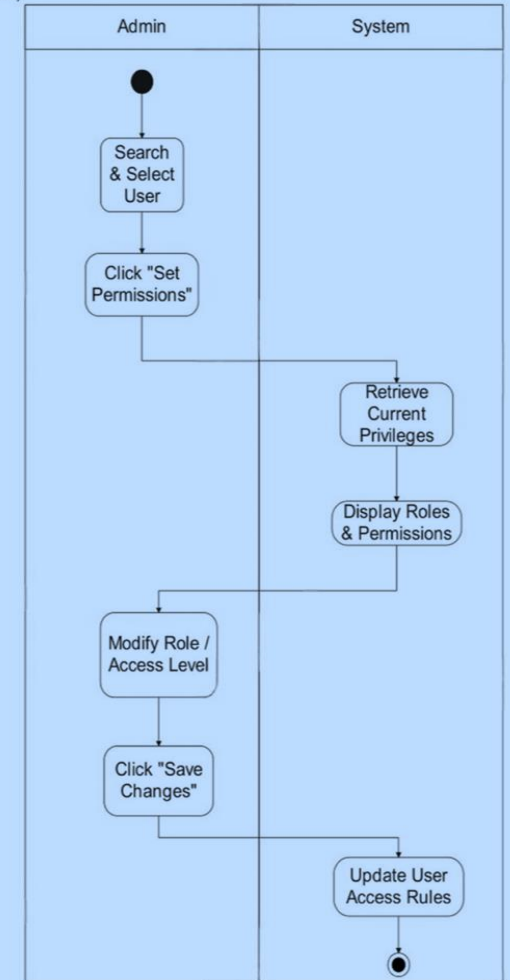


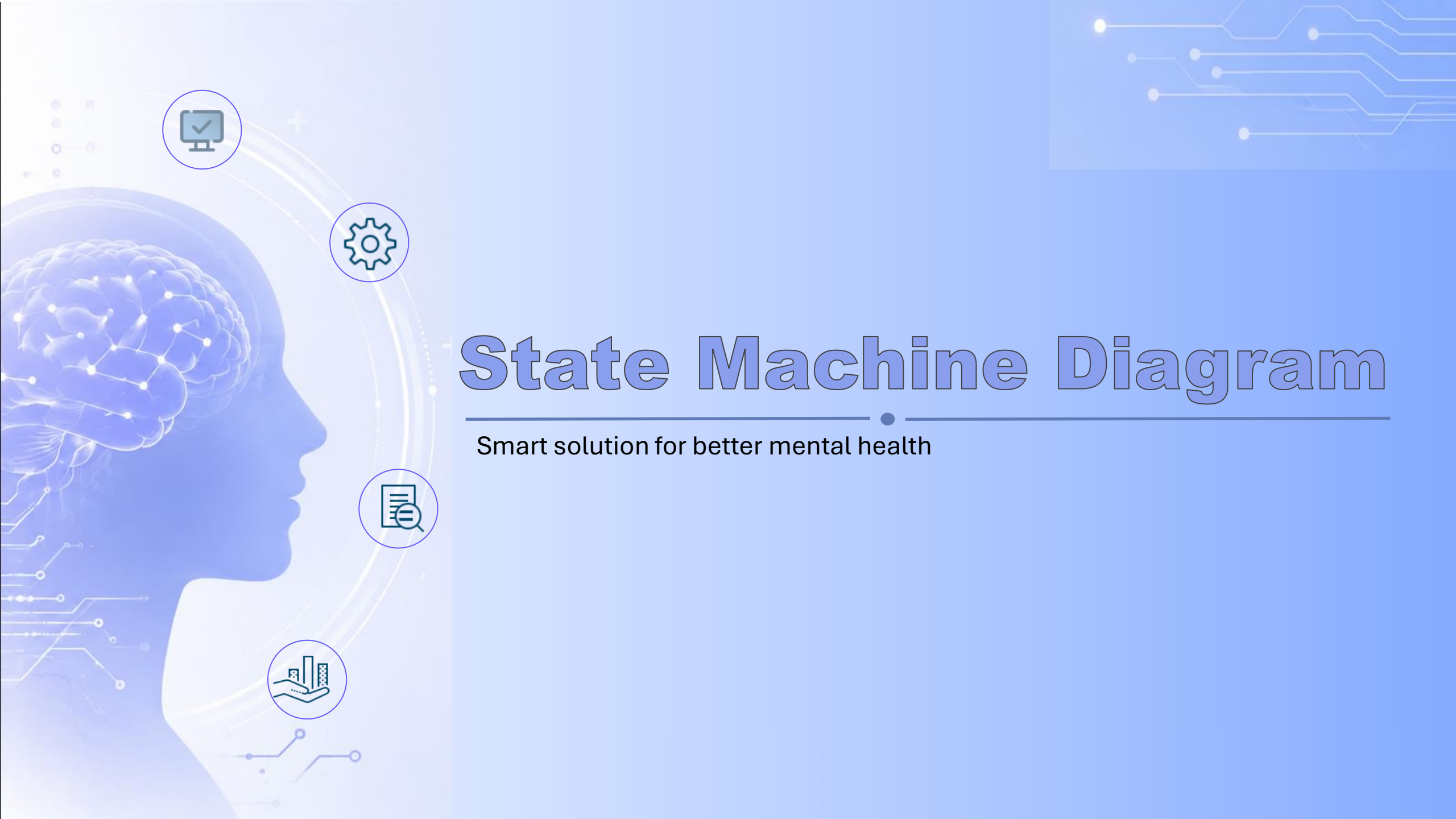


Activity Diagram  
For Use Case :  
(Manage Users)



Activity Diagram  
For Use Case :  
(Set User  
Permissions)





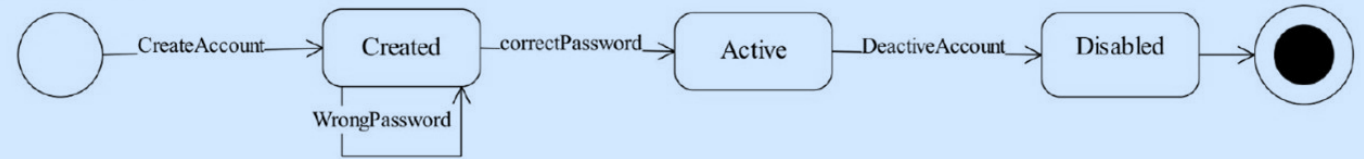
# State Machine Diagram

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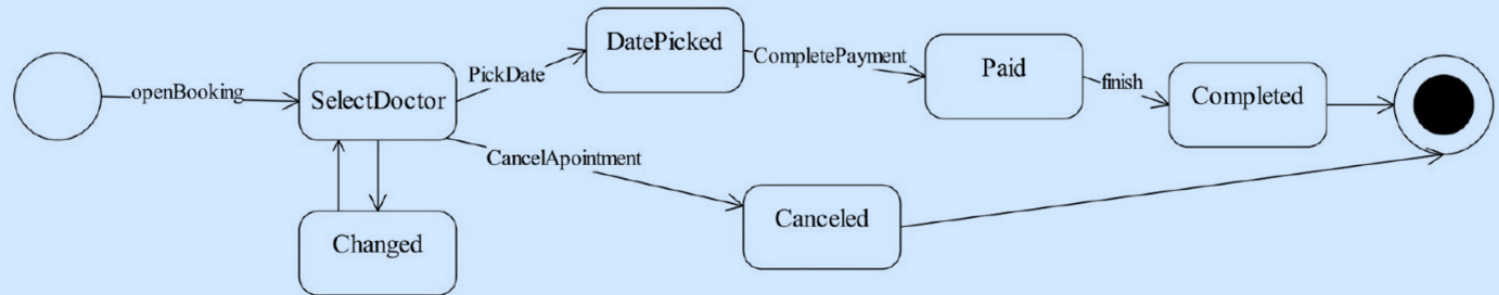
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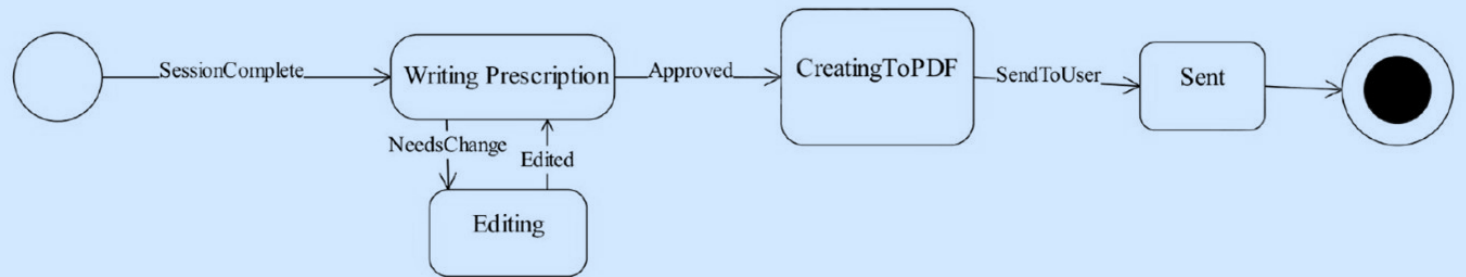
### USER STATE



### USER BOKING



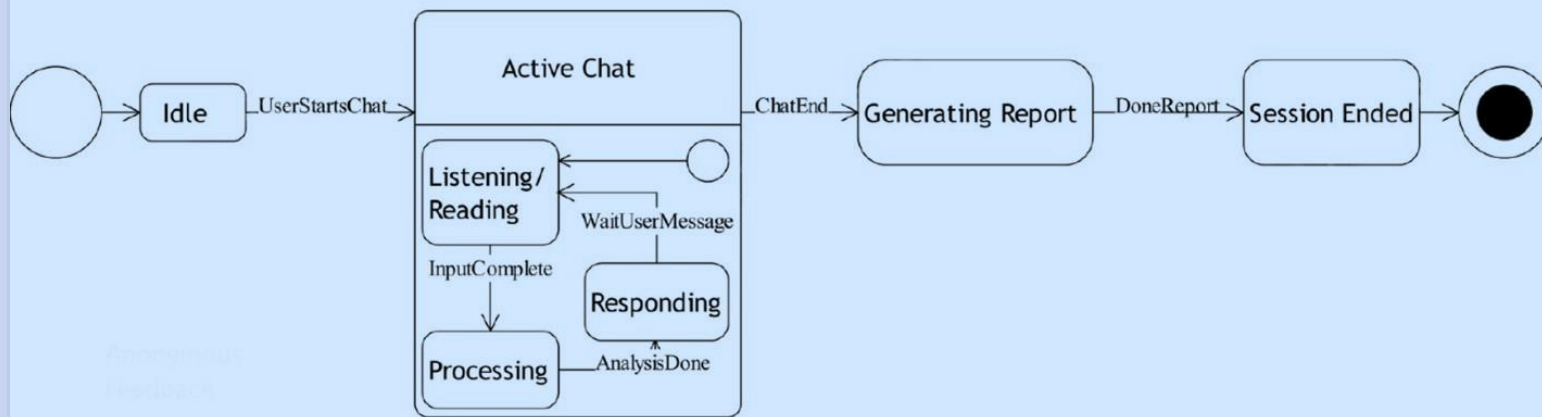
### PRESCRIPTION



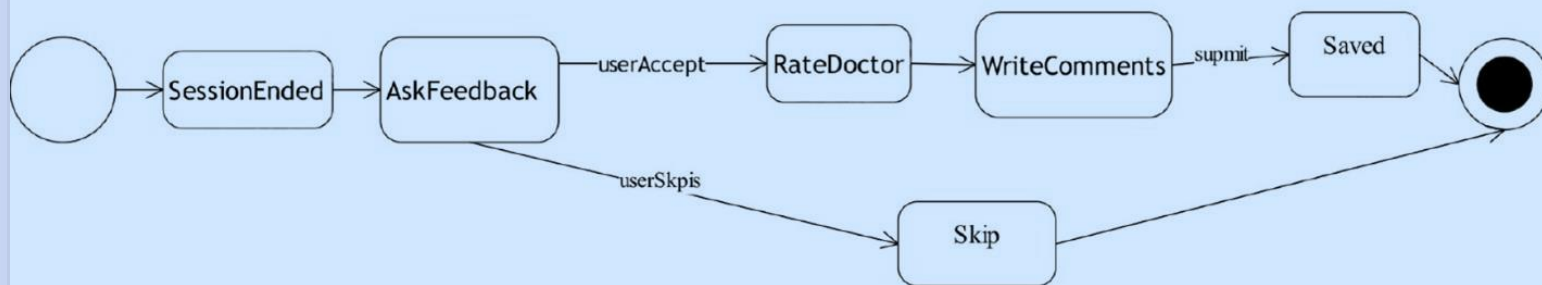




## AI INTERACTION



## ANONYMOUS FEEDBACK

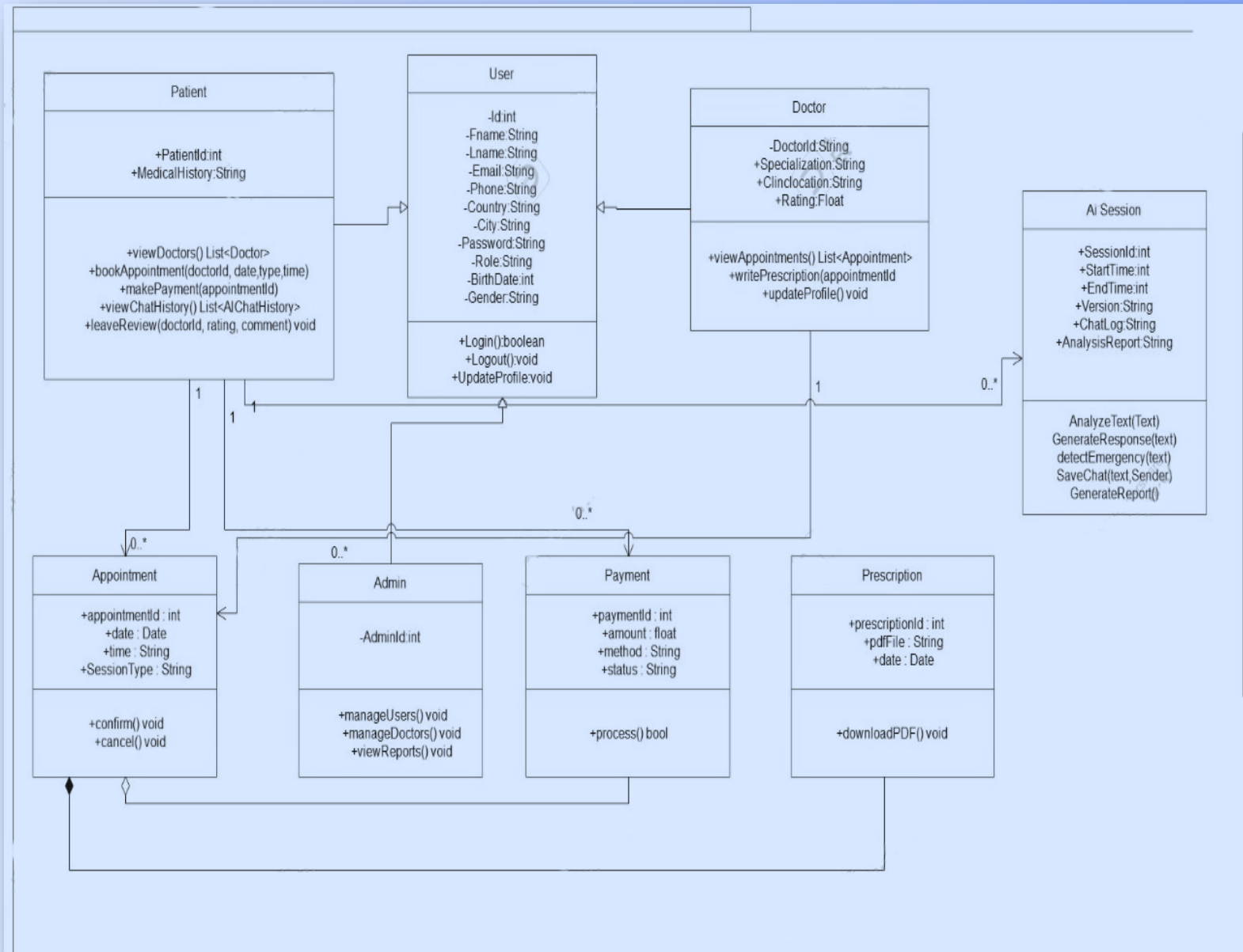




# Class Diagram

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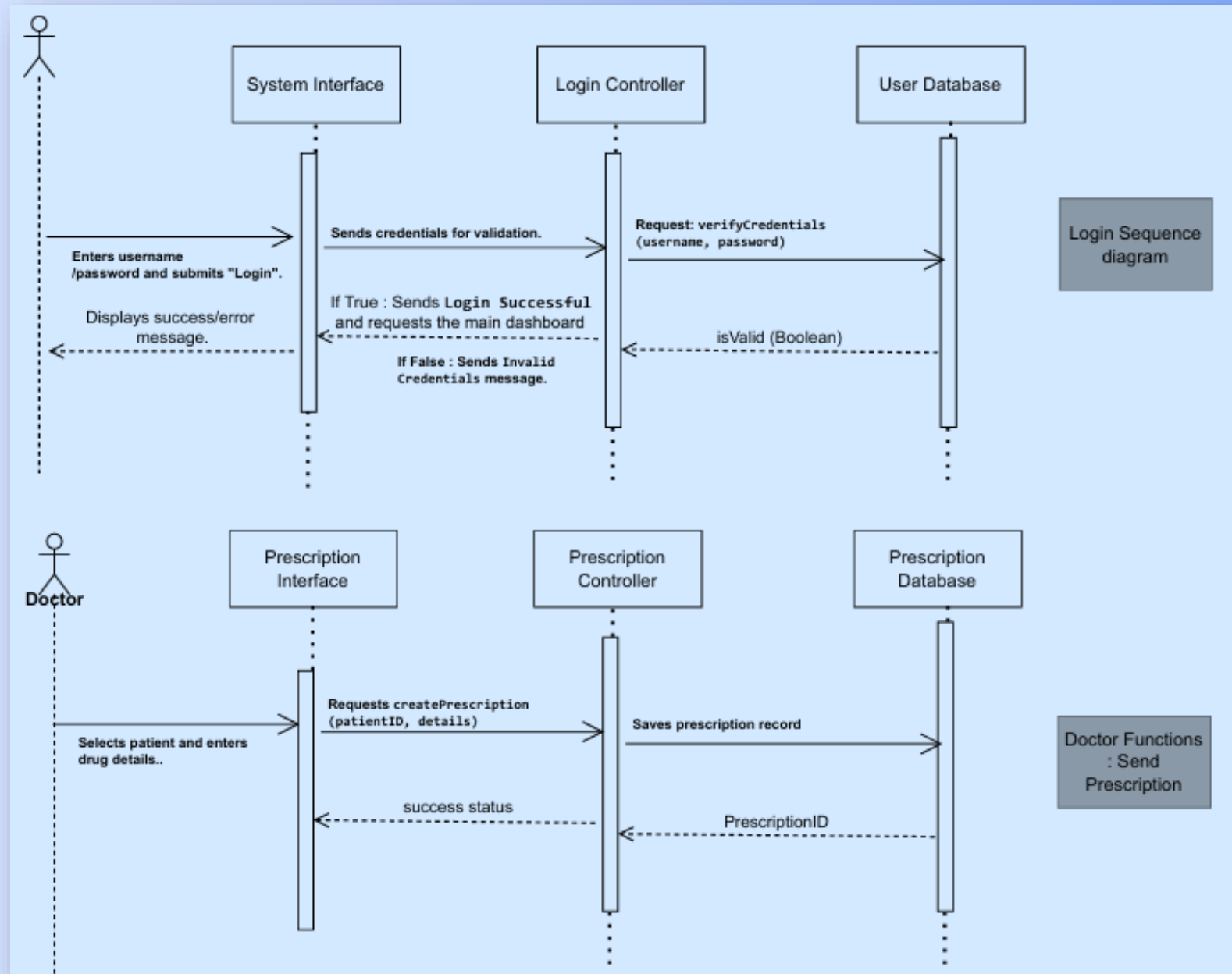


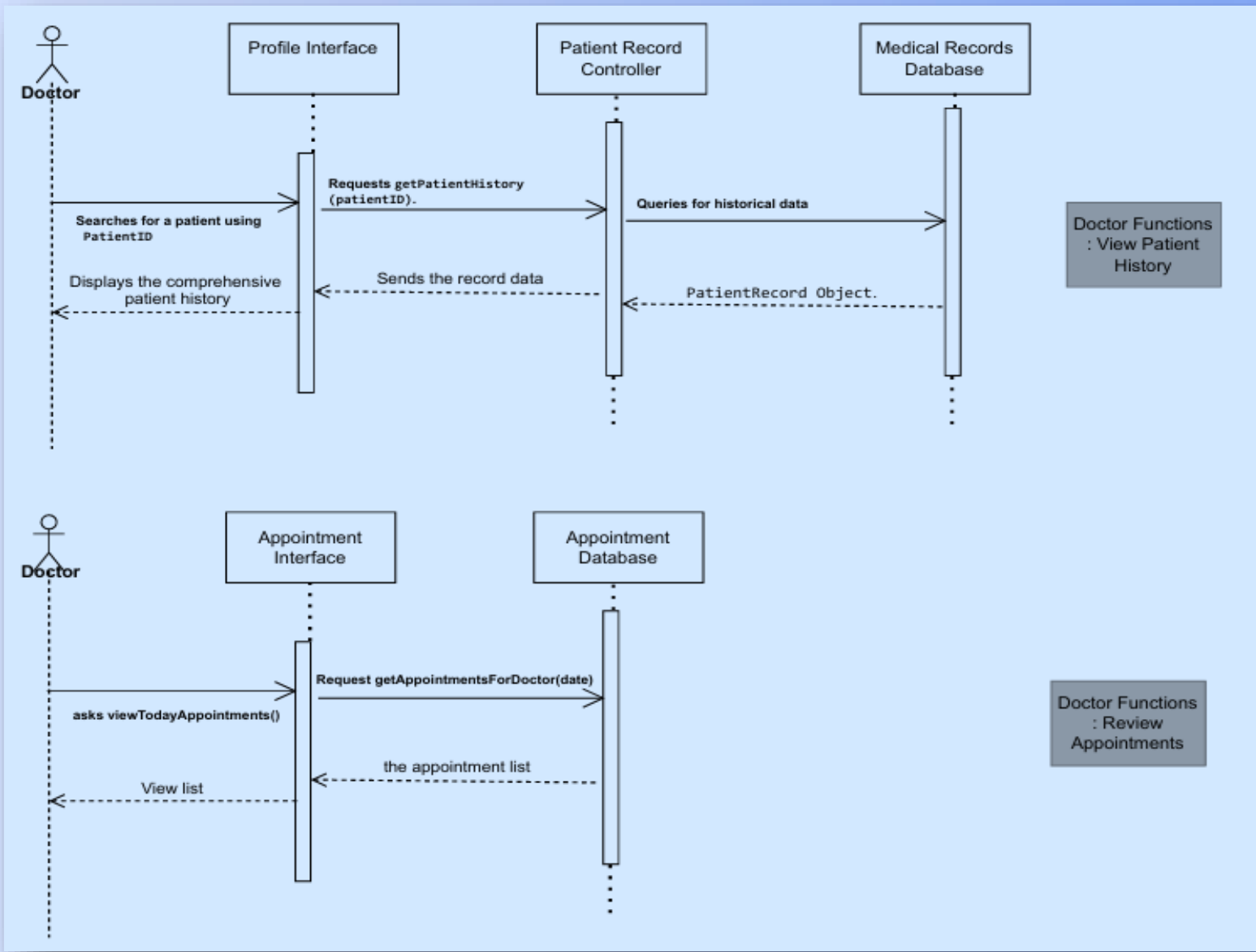


# Sequence Diagram

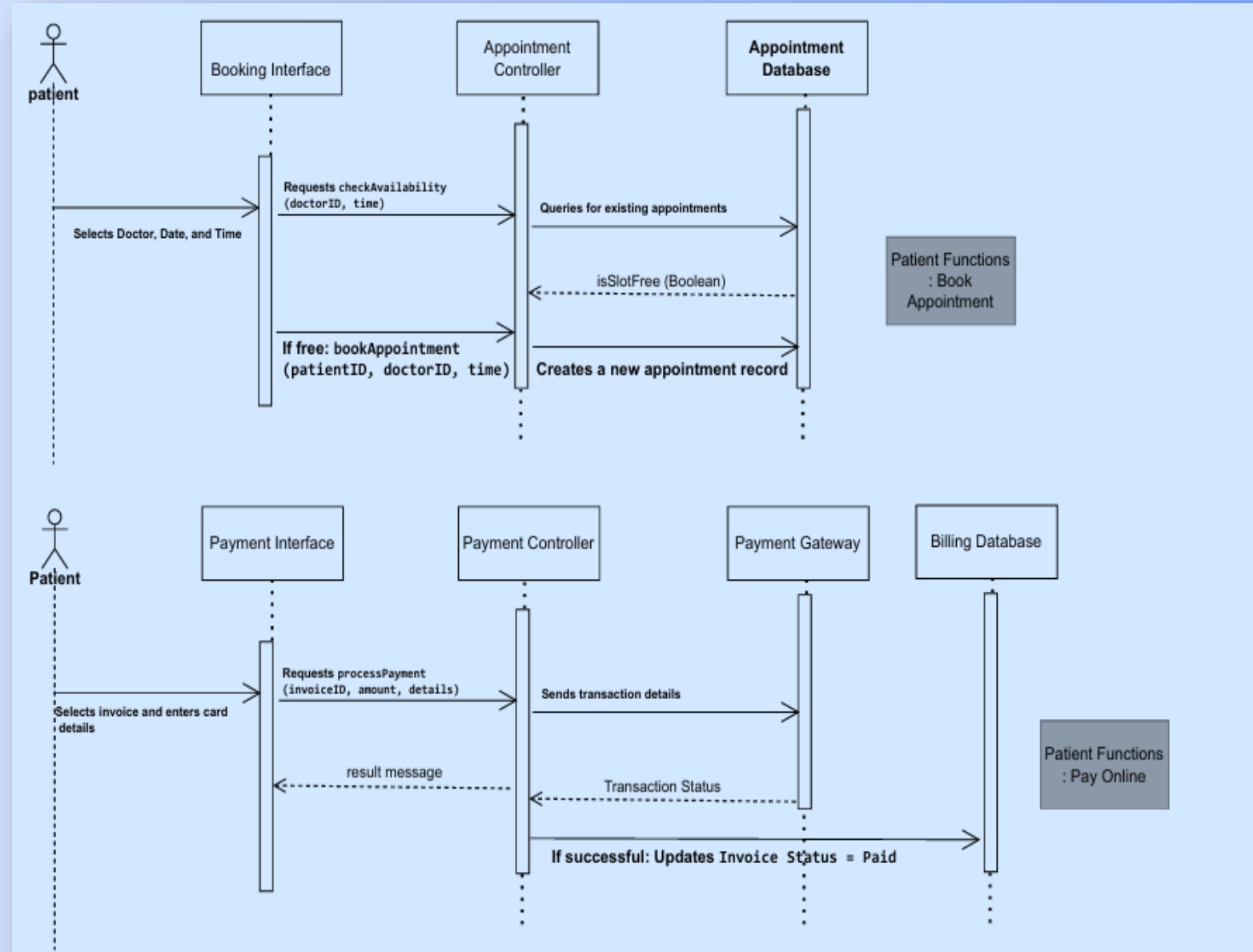
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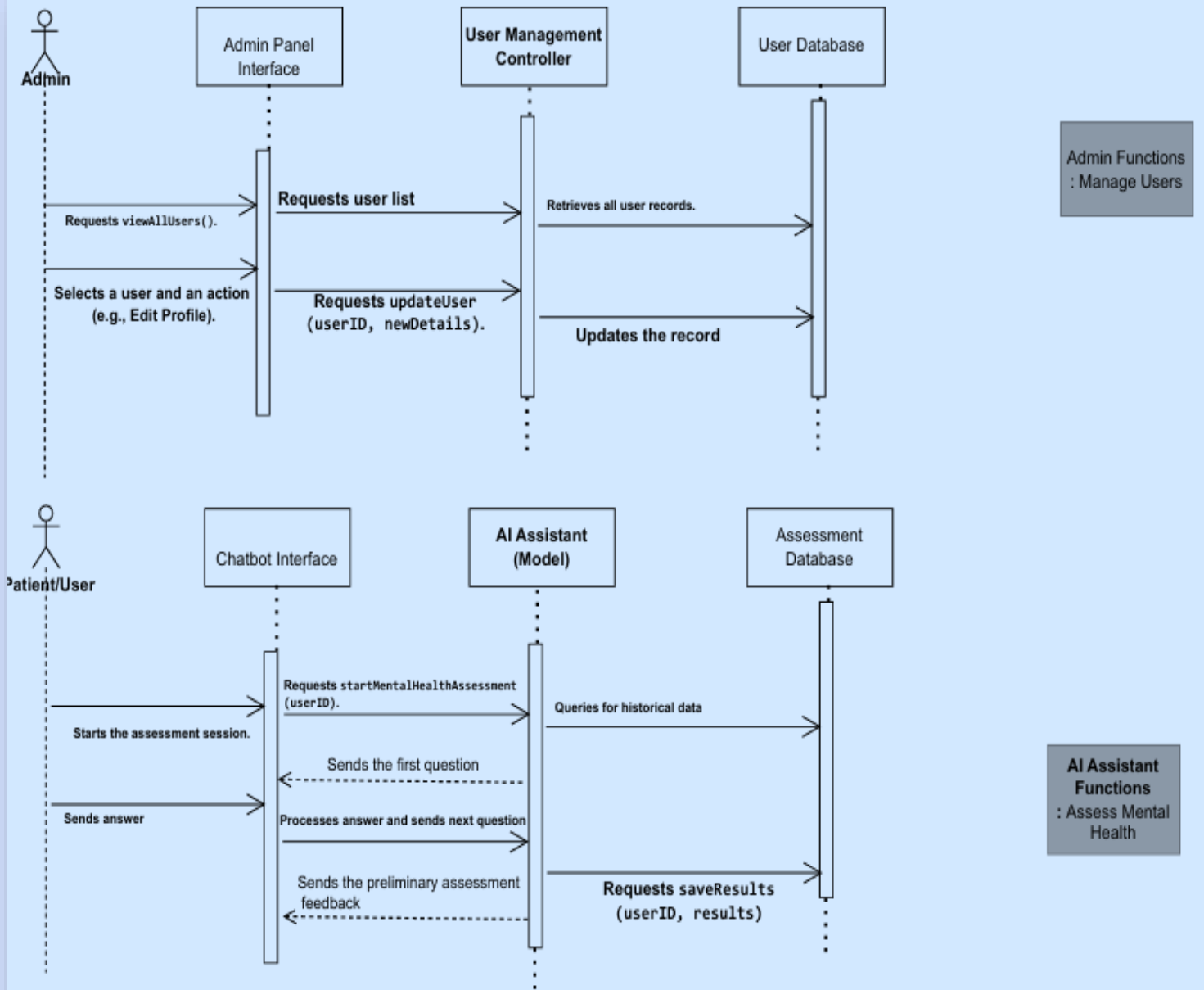
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# Implementation

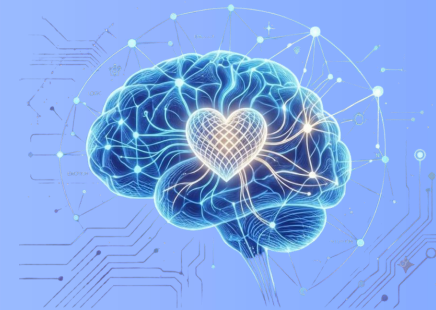
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# Implementation Technologies



## Frontend

**HTML, CSS, and JavaScript:** for building responsive and interactive user interfaces.

**React.js:** for developing reusable components such as dashboards, and appointment management interfaces .

**Ready Player Me:** for creating personalized 3D avatars.

**Three.js:** for rendering and displaying interactive 3D avatars within the platform.



## Backend

**ASP.NET Core Web API:** for secure APIs and system services .

**Onion Architecture:** for a scalable and maintainable structure .

**Entity Framework Core & LINQ** for data access and querying .

**SQL:** for managing user accounts, profiles appointments, and medical records.

**JWT Authentication:** for secure user login

**Flask API Integration** for connecting AI-based mental health analysis services.



## Tools

**Git & GitHub** – Version control and team collaboration .

**Postman** – API testing and validation .

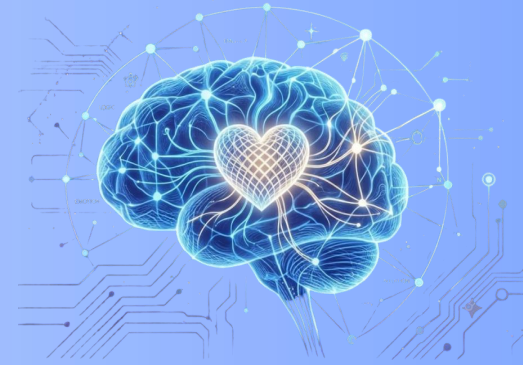
**Swagger UI** – API documentation and endpoint testing .

**Visual Studio** – Backend development and debugging .

**Visual Studio Code** – Frontend development and code editing.



# Key Systems Modules



## 1 User Management

### **Purpose:**

Allows users to create accounts, log in securely, and manage their profiles .

### **Technology:**

JWT for authentication .  
bcrypt for password encryption .  
SQL Server for storing user data .  
React Forms for registration and login pages.

## 2 AI Chatbot & Virtual Robot

### **Purpose:**

Provides an initial mental health assessment and interacts with users through conversations .

### **Technology:**

Scikit-learn models for mental health analysis .  
Emotion Detection Through Voice Analysis .  
React-based chatbot interface.

## 3 Appointment Booking

### **Purpose:**

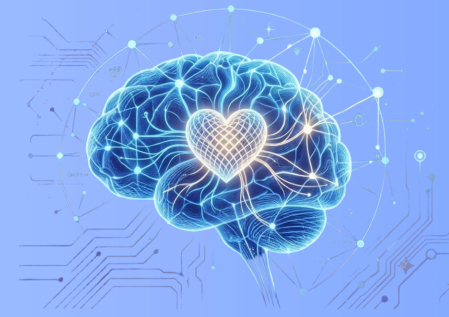
Enables patients to schedule and manage online or offline therapy sessions .

### **Technology:**

ASP.NET Core APIs .  
React Calendar Components.



# Challenges and Solutions



## AI Accuracy in Emotional Assessment

Trained on diverse datasets with continuous validation and model improvement.

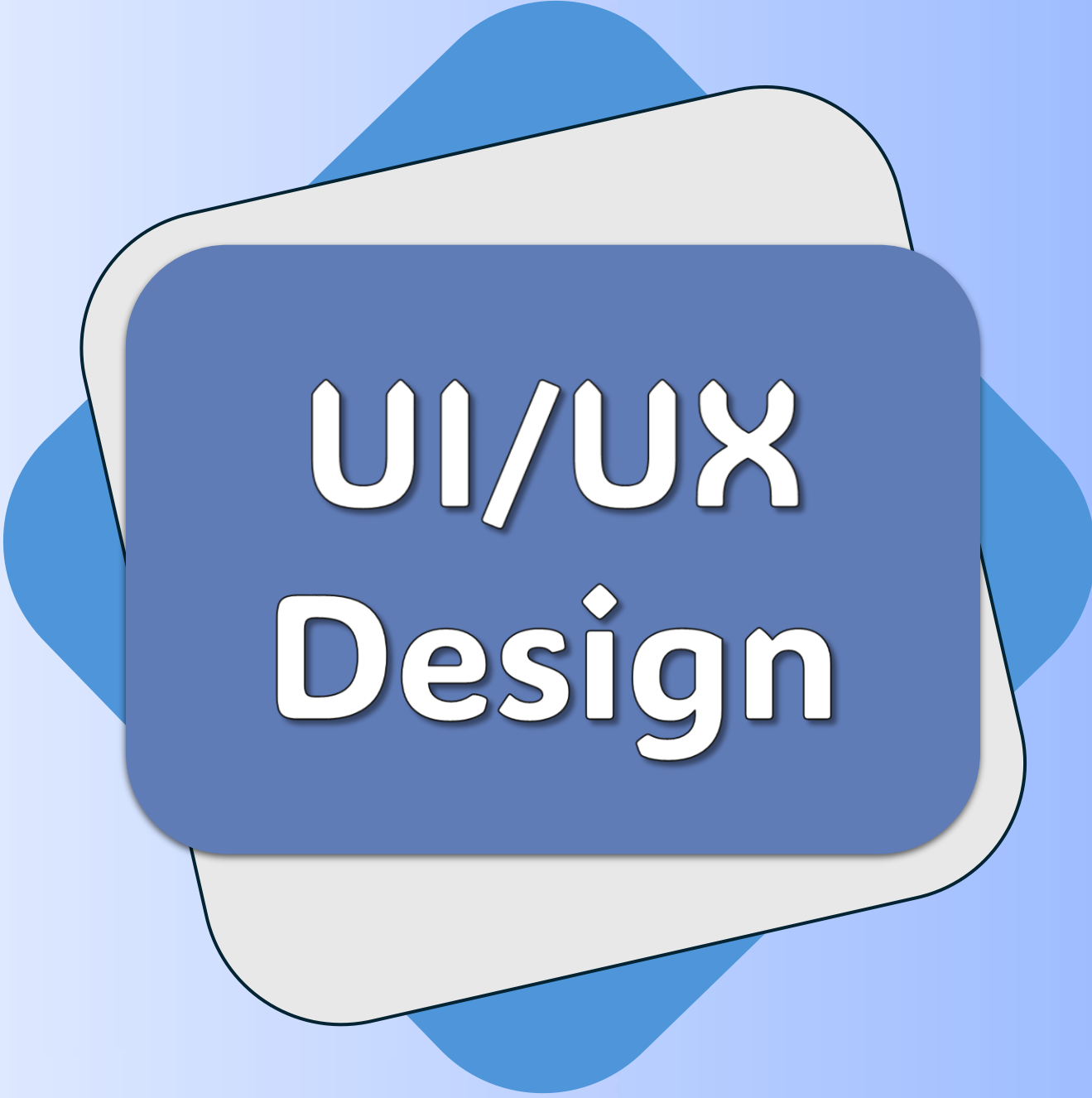
## Privacy & Sensitive Data Security

Encrypted database and role-based access control (RBAC).

## System Scalability

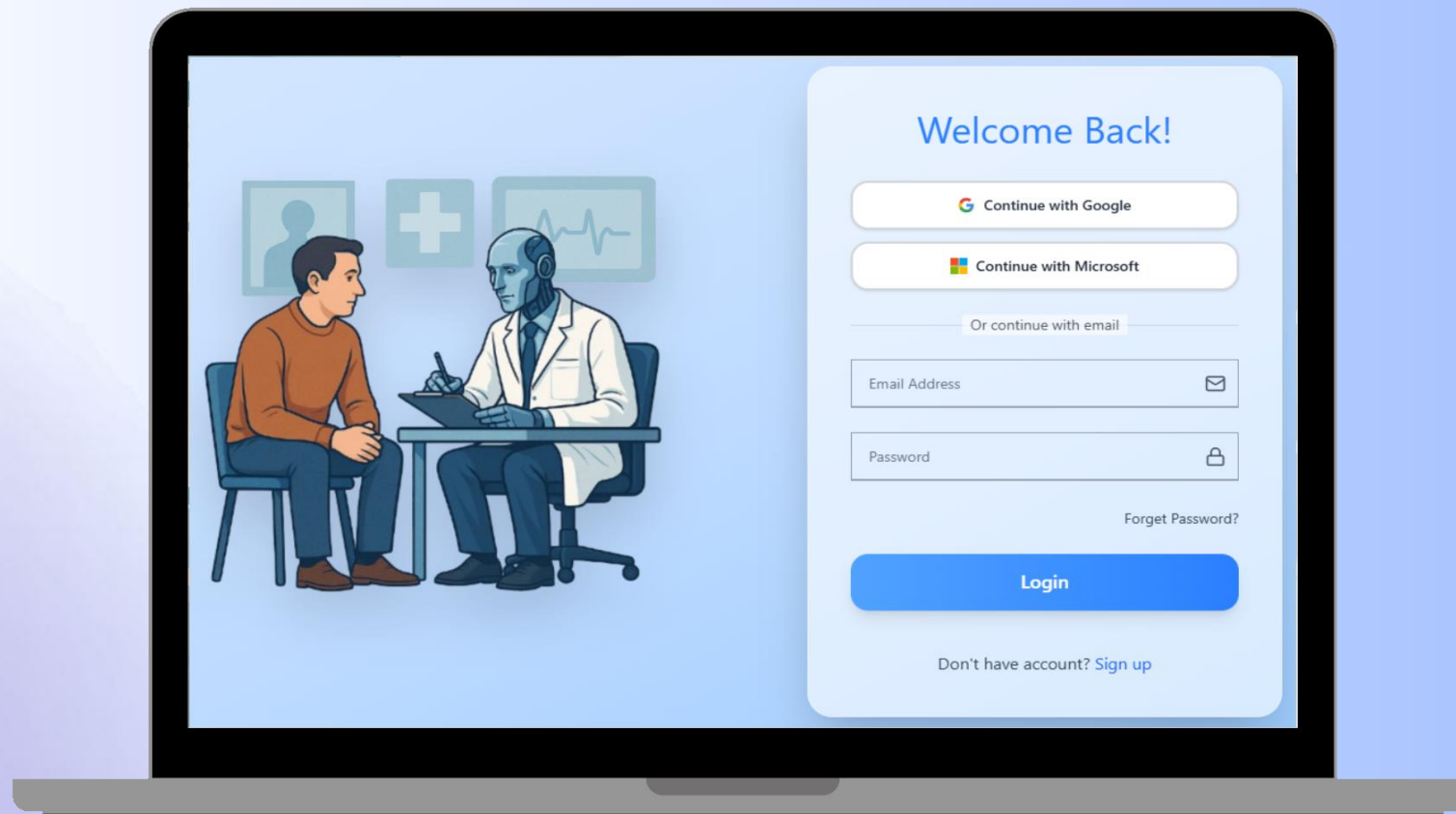
Onion Architecture with ASP.NET Core for a scalable and maintainable system



The graphic features a central blue rounded rectangle with the text 'UI/UX Design' in white. This rectangle is layered on top of a light gray rounded rectangle, which is in turn layered on top of a blue rounded rectangle. The background is a light blue gradient with a soft purple and white gradient on the left side.

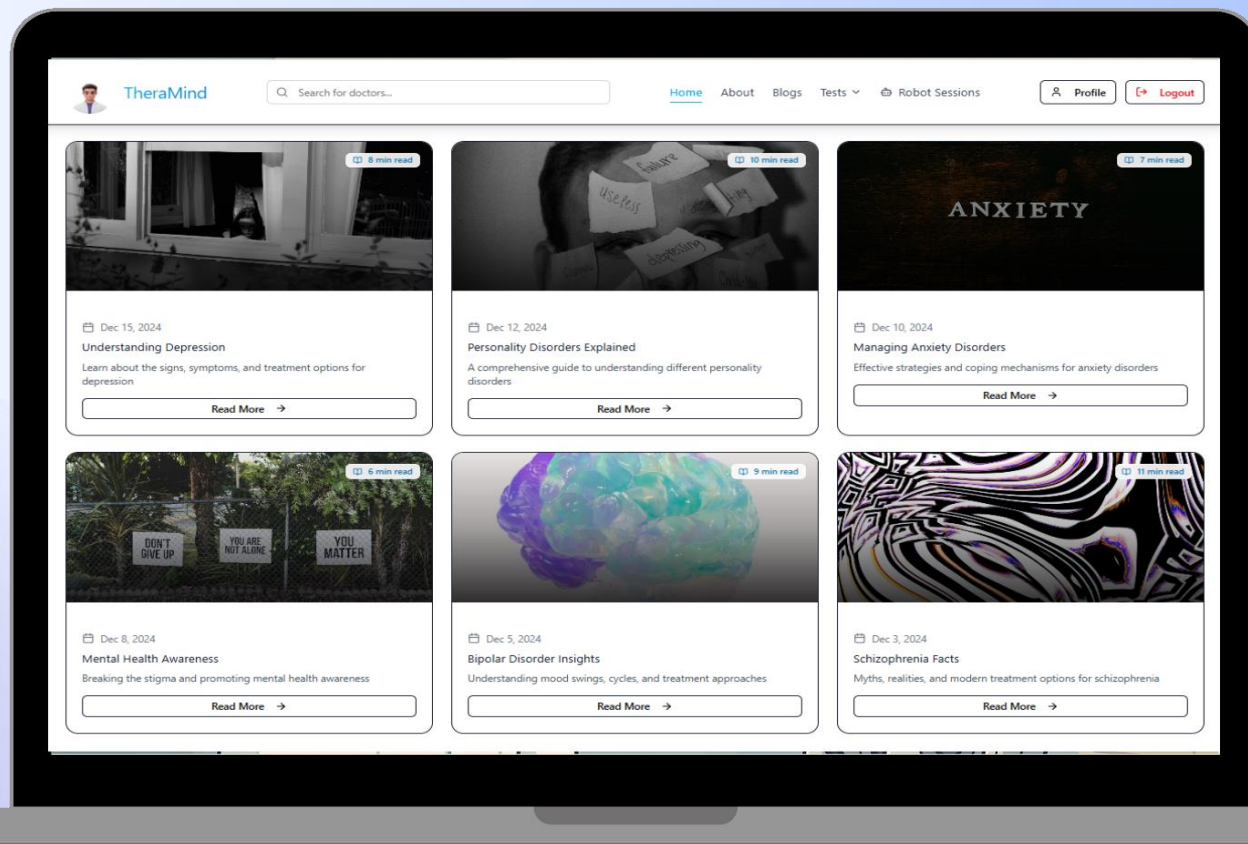
# UI/UX Design

# Login page



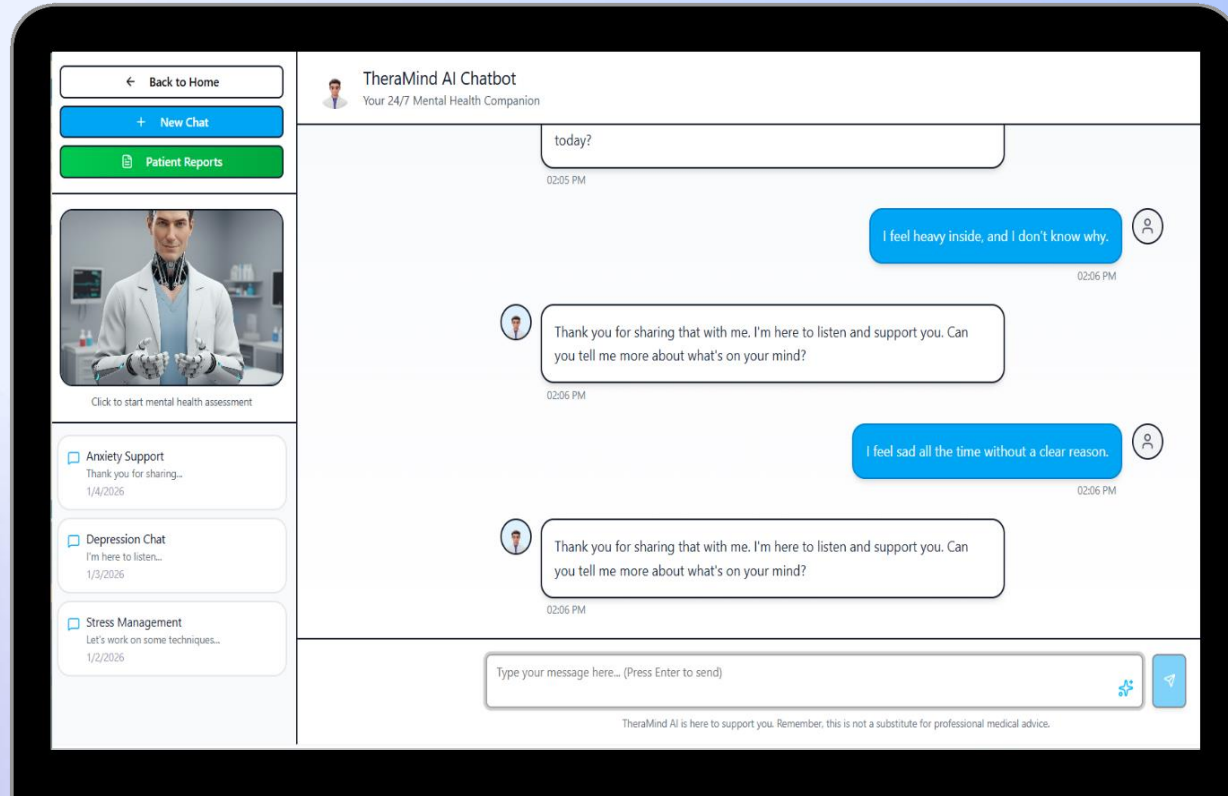


# Home page





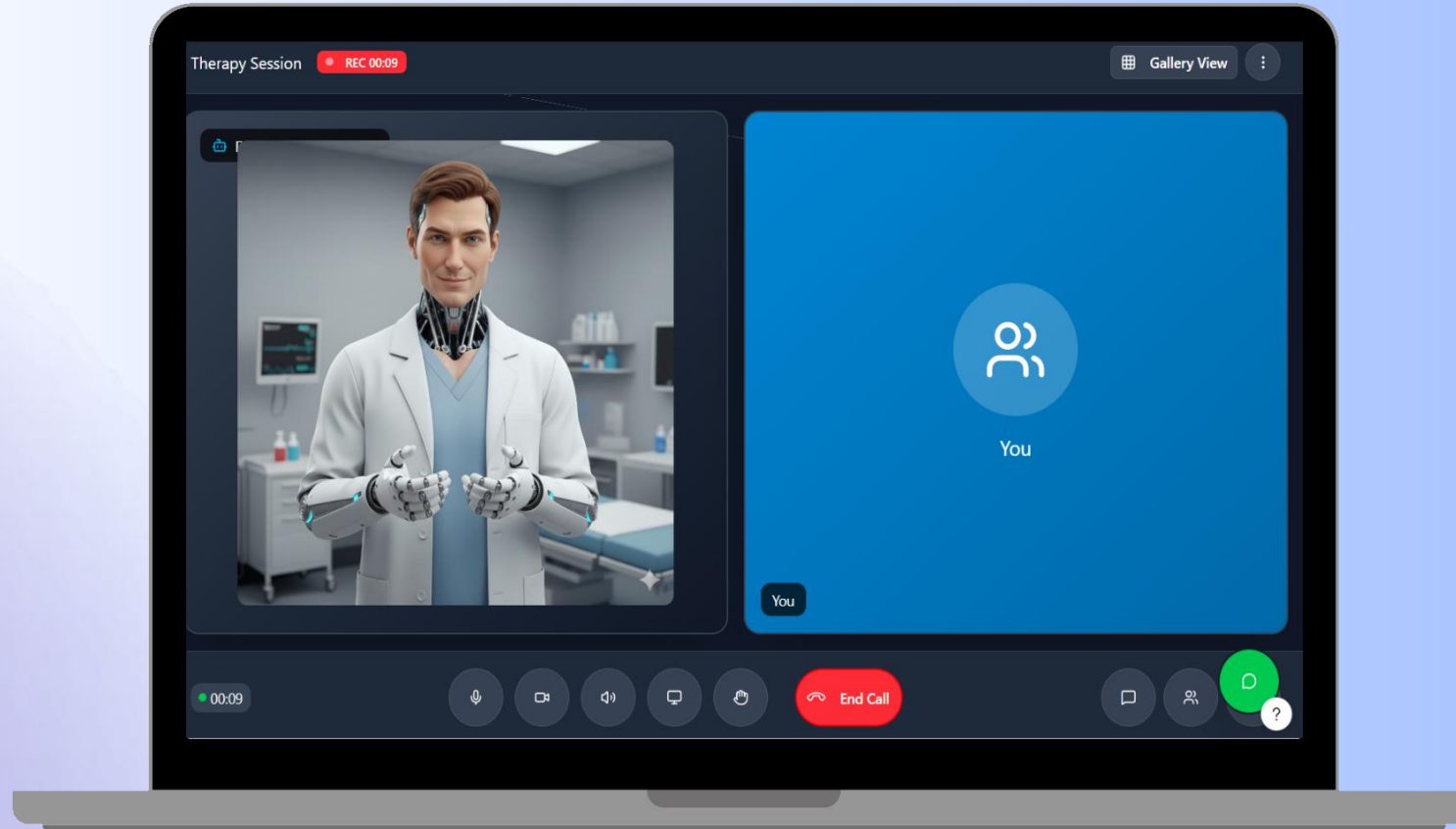
# Chatbot page



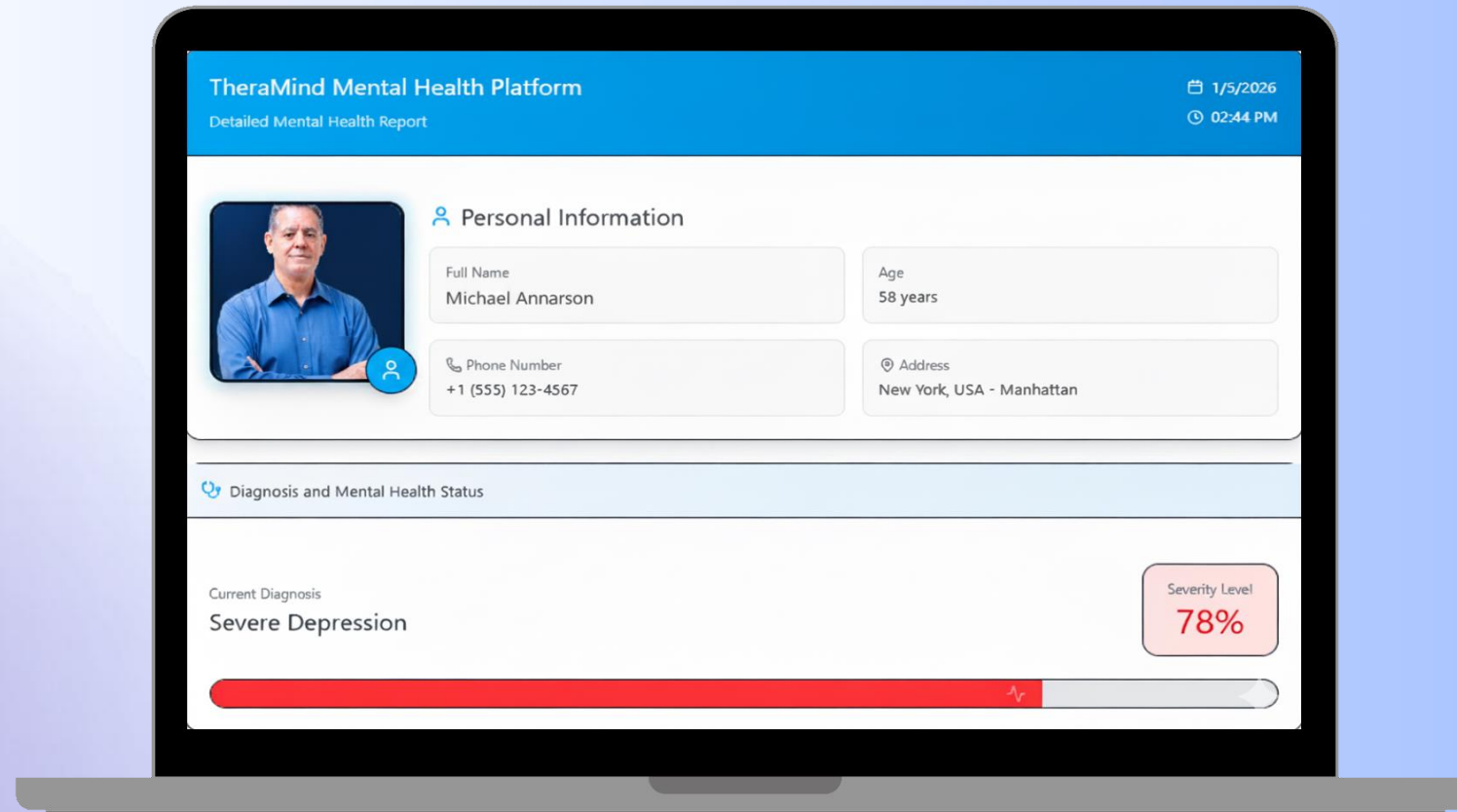




# Robot meeting page



# Report page





# Doctor profile page





**Dr. John Smith**  
Clinical Psychologist & Therapist  
Cairo, Egypt 500+ Patients 15 Years Experience

Settings Prescription Book Appointment

**Overall Rating**  
**4.9** ★★★★★  
Based on 342 reviews

|         |             |     |
|---------|-------------|-----|
| 5 stars | <div></div> | 85% |
| 4 stars | <div></div> | 10% |
| 3 stars | <div></div> | 3%  |
| 2 stars | <div></div> | 1%  |
| 1 stars | <div></div> | 1%  |

**About Dr. John Smith**

Dr. John Smith is a highly experienced Clinical Psychologist with over 15 years of experience in mental health care. He specializes in treating depression, anxiety disorders, PTSD, and provides cognitive behavioral therapy.

Dr. Smith graduated from Cairo University Faculty of Medicine and completed his specialization in Clinical Psychology at the American University in Cairo. He has helped over 500 patients overcome their mental health challenges through evidence-based therapeutic approaches.

His patient-centered approach focuses on building trust, understanding individual needs, and creating personalized treatment plans that deliver real results.

**Specializations**

Depression Anxiety Disorders PTSD Cognitive Therapy

Family Counseling Stress Management

**Services Offered**

 **Online Sessions**  
Video consultations from anywhere  
\$80/session

 **In-Clinic Sessions**  
Face-to-face consultations  
\$100/session

 **Chat Support**  
Text-based therapy sessions  
\$60/session

 **Group Therapy**  
Small group sessions  
\$40/session

**Availability**

Mon - Fri: 9:00 AM - 8:00 PM

Sat: 10:00 AM - 4:00 PM

Sun: Closed



# Doctor booking page

**TheraMind**

[Home](#) [About](#) [Blogs](#) [Tests](#) [Robot Sessions](#) [Profile](#) [Logout](#)

## Book Your Appointment

**Session Type**

☒  **Online Session**  
Video consultation  
\$80

☐  **In-Clinic Session**  
Face-to-face  
\$100

**Select Date & Time**

**Select Date**



**Select Time**

09:00 AM

10:00 AM

11:00 AM

12:00 PM

02:00 PM

03:00 PM

04:00 PM

05:00 PM

06:00 PM

07:00 PM

### Summary

Type [Online](#)

Total \$80

[Join Meeting](#)

Click to start your video consultation







# Doctor review page



**TheraMind**[Home](#)[About](#)[Blogs](#)[Tests](#)[Robot Sessions](#)[Profile](#)

### Rate Your Session

How was your experience with Dr. Yahia Hassan?

Overall Rating \*



Very Good

How would you rate the session quality? \*

☐ Excellent - Exceeded expectations

☒ Good - Met expectations

☐ Average - Satisfactory

☐ Poor - Below expectations



# Testing and Evaluation

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Smart solution for better mental health



# Unit Testing

**Goal:** Validate Individual Functions in Isolation

**Tools:** xUnit / NUnit (Backend) — React Testing Library (Frontend)



## Frontend

- 1) Testing of AI Chatbot UI components and message rendering
  - 2) Validation of login and registration forms
  - 3) Doctor profile component rendering
  - 4) Appointment booking form validation
  - 5) Virtual AI Avatar interface  
3D avatar UI, speech display, emotion response animations
- 



## Backend

- 1) Authentication services (login/register and JWT generation)
  - 2) AI psychological assessment logic
  - 3) Appointment scheduling functions
  - 4) Medical record creation and retrieval APIs
  - 5) Virtual AI assistant response processing and emotion analysis
-



# Integration testing

**Goal:** Ensure Seamless Interaction Between Modules via RESTful APIs

**Tools:** Postman, Swagger UI

## Frontend ↔ Backend

React Frontend communication  
with ASP.NET Core APIs

## AI Chatbot & Avatar

AI Chatbot and Avatar  
Assessment Processing

## Medical Records

Storage and Retrieval  
in SQL Server

## Appointment Workflow

Availability Check → Booking  
→ Confirmation



## Test Scenario

User Login → AI Chatbot Session → Psychological Report → Doctor Recommendation → Appointment Booking

## Result

Smooth Data Flow Across All Modules





# User testing

**Goal:** Ensure Seamless Interaction Between Modules via RESTful APIs

**Tools:** Postman, Swagger UI

## Participants

Students  
Volunteers

## Tested Features

Chatbot  
Avatar  
Assessment  
Booking

## Feedback Methods

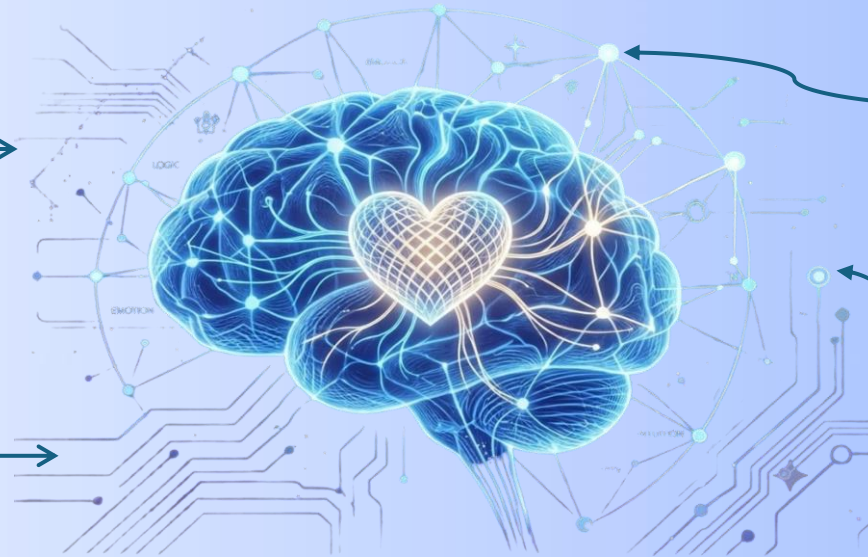
Surveys  
Observation

## Results

Realistic Avatar  
Supportive AI Chatbot  
Positive User Experience  
High User Engagement



# System Evaluation



## Digital psychiatrist

**150–250 ms**

### API RESPONSE TIME

Averaged for all core features  
(tested via Postman and Swagger UI)

**Stable**

### CONCURRENT USERS

System handled multiple simultaneous  
AI interactions and appointment requests  
without performance issues

**< 2 s**

### FRONTEND LOAD TIME

Kept under 2 seconds using  
React lazy loading and code splitting

**85%+**

### USER SATISFACTION

of testers rated the platform as  
"easy to use, fast, and emotionally  
supportive"

### ACCURACY

- AI correctly detects depression indicators.
- Generates accurate psychological assessments.

### SPEED

- Lazy loading in React.
- SQL Server indexing.
- Caching for faster performance.

### SCALABILITY

- Horizontal scaling.
- Optimized for large data.
- AI module ready for future upgrades.

## Why Our Platform Stands Out

### AI GUIDANCE

**Our Platform:** AI guidance before consultation.  
**Others:** No AI guidance.

### PROFESSIONAL CARE

**Our Platform:** Connects you with real psychiatrists.  
**Others:** Chat only.

### SMART BOOKING

**Our Platform:** Integrated appointment system.  
**Others:** Limited or manual booking.



# Future Work

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Smart solution for better mental health

01



## AI-Powered Emotion Analysis

Detects emotional patterns through facial expressions to support early depression detection.

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02



## Doctor Recommendation

Suggests the most suitable doctors based on patient needs , reviews, and location .

---

03



## Secure online payment

Enables fast and secure electronic payments after appointment booking.

---

04



## Digital Prescriptions

Allows doctors to issue secure digital prescriptions during online consultations .

---

05



## Anonymous Doctor Review

Lets patients rate and review doctors anonymously after each session .

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**Thank you**